

Crisis Bargaining Over What?

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Abstract

Using a game-free approach, I study crisis bargaining games with uncertainty about the common value of the contested object, or *stakes*. I show that, under mild conditions, there is no crisis bargaining game with uncertain stakes in which transfers can remedy a bargaining failure caused by an indivisibility of the object. This impossibility arises because uncertainty about stakes breaks the equivalence between allocating shares of the object and exchanging transfers, such as side payments or concessions on other issues. This is not true for the standard sources of uncertainty about power and the costs of war, which preserve that equivalence. Whether states are unable to reach an agreement or maintain it depends on the location of the indivisibility. I also show that correlated information can exacerbate the problem by amplifying demands when stakes are thought to be high.

Keywords: conflict, bargaining, information, game theory, mechanism design

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1 Introduction

Throughout the nineteenth century, the United States repeatedly bought large territories from its international rivals without resorting to war.¹ France sold Louisiana in 1803 for \$15 million, Spain ceded Florida in 1819 in exchange for the assumption of \$5 million in claims against it, and Russia sold Alaska in 1867 for \$7.2 million. Yet, when US President Polk dispatched an envoy to Mexico City in 1845 who was authorized to offer as much as \$25 million for New Mexico and California, the mission collapsed and war followed within months. Why was peaceful agreement accomplished in Louisiana, Florida, and Alaska, but not in California? Was there a failure of the diplomatic process, or was there no mechanism for resolution that could be sure to avoid war?

The core difference between these cases is that (i) California’s value was highly uncertain, and (ii) a voluntary cession of territory was politically infeasible for the Mexican government. First, neither government knew the extent of material wealth that came with control of California, though there was reason to suspect its value could be immense. In May 1846, the American consul at Monterey reported to the Secretary of State that “gold, silver, quicksilver, copper, lead, sulfur, and coal mines are to be found all over California” (Hammond, 1953). Exactly how much wealth the territory would bring was unclear. Second, Mexican governments at the time were fragile, and openness to ceding territory to the US was viewed as being tantamount to treason. Mexican President Herrera was deposed in December 1845 amid such accusations for his willingness to receive Polk’s envoy, likely providing his successors with a lesson on the consequences of any cooperation with American expansion. Henderson (2007) provides a historical account, contending that Mexican leaders came to prefer war they expected to lose to voluntarily giving up territory.

This paper shows that these two features make it impossible to consistently avoid war. When a contested object cannot be divided, the standard remedy is transfers, usually in the form of side payments or concessions on other issues. Transfers have widely been thought to render any object effectively divisible (Fearon, 1995, p. 389). Jackson and Morelli (2007, p. 1356) state the claim plainly: “if one allows for bargaining and transfers, war should not be possible.” This reasoning is common in the literature, dating back to at least Morrow (1986) and Morgan (1990). However, this remedy relies on an equivalence between a share of the object and a transfer worth the amount of that share, which no longer holds with uncertainty about the contested object’s value. A share

¹ For a full historical account, see Herring (2008).

of the object is worth more when the object proves valuable, whereas a transfer’s value is not tied to the object directly and cannot be denominated in units of an unknown quantity.² In the case of California, neither the US nor Mexico knew what the territory was worth, so no level of compensation to either side could have reliably substituted for the divisions that Mexico could not concede.

The results apply to every crisis bargaining game. Accordingly, the analysis does not hinge on the assumptions of a particular setting. To accomplish this, I employ a “game-free” approach that leverages tools from Bayesian mechanism design that have proven useful for studying information problems in war due to their ability to facilitate general statements about broad classes of games (Banks, 1990; Fey and Ramsay, 2009, 2011). The focus is on war and peace as possible equilibrium outcomes while remaining agnostic about the structure of the game, such as actions and timing. Of special interest is the existence of a crisis bargaining game that admits an *always-peaceful equilibrium* in which war never occurs on the path of play, and whether transfers can help restore always-peace when some divisions of the object are not available as settlements. I refer to any such constraint on feasible divisions of the object’s value as an *indivisibility*, which (as defined here) can stem from any source, such as characteristics of the disputed object, domestic political constraints, institutional limitations of the resolution process, or other features of the strategic interaction.

Importantly, the analysis does *not* require indivisibilities in the usual payoff-based sense, as no restriction is placed on the payoffs states can attain in peaceful settlements (which indivisibilities usually imply throughout existing work, where the stakes of the dispute are common knowledge). Because transfers are left unconstrained throughout the analysis, every division of the surplus from peace remains attainable at any of the feasible divisions of the object. Thus, an indivisibility here refers to any restriction on settling peacefully at some divisions of the object’s *value*, not necessarily of the literal object.³ Moreover, the object’s common value implies that every division yields the same surplus, so war results from private information alone rather than from any problem of allocating the object efficiently.

The central takeaway of the paper is that, contrary to conventional wisdom, there is no crisis bargaining game in which transfers can remedy a bargaining failure caused by an indivisibility when

² This is definitional and not an assumption about contracting. A payment indexed to the realized value of the object would constitute a share of it, and is therefore considered a division of the object, not a transfer. For example, an agreement promising Mexico a percentage of the profits from California’s mines is a share of the mine’s value.

³ A literal inability to divide the object does not necessarily imply an indivisibility in its value if states can credibly commit each side to a portion of what the object yields (e.g., a profit-sharing scheme).

there is uncertainty about stakes.⁴ In fact, neither uncertainty about power nor uncertainty about the costs of war has this consequence. When the stakes of the dispute are known, transfers can always provide compensation for concessions on the division of the object, so the probabilities of war attainable across all crisis bargaining games are the same with or without indivisibilities. The substitution of transfers for shares does not require information beyond the value of the contested object, so it holds regardless of any uncertainty about power or costs of war that may persist (Proposition 1). Further, when every division of the object’s value is feasible, there always exists a crisis bargaining game in which states can settle by allocating shares (Remark 1).

Theorem 1 establishes the impossibility result for boundary indivisibilities, i.e., situations in which the feasible set of peaceful settlements excludes shares to an extreme. When “fair” shares that would guarantee peace are no longer feasible, states must rely on transfers to bridge the gap between feasible shares and their privately held expectations about the object’s value. However, states then face a fundamental tension: if one state offers a large transfer to compensate an opponent that claims the stakes are high, then an opponent with low expectations has an incentive to make the same demand. Anticipating this, the extending state will not concede transfers large enough to always satisfy states with high expectations. No transfer scheme can resolve this tension and, as a result, provided the costs of war are not too large, war occurs with positive probability in every equilibrium of every crisis bargaining game with such an indivisibility (Theorem 1).

I also consider an interior indivisibility, in which extreme divisions remain available but some intermediate divisions are infeasible.⁵ In this case, there are crisis bargaining games in which peace is always reached (Proposition 3) but none in which it is always maintained (Theorem 2). Indeed, no robust equilibrium of any such game is always peaceful, despite the possibility of unconstrained transfers that could, in principle, help states reach a mutually preferred settlement. Even though transfers could remedy the problem caused by the indivisibility in principle, in practice, each side has an incentive to misrepresent their private information about the stakes of the dispute in attempt to receive a more favorable package of shares and transfers.

⁴ As with any impossibility result in crisis bargaining, this conclusion requires that the costs of war not be so large that peace becomes trivially preferred (e.g., Fey and Ramsay (2011)). I largely suppress this inevitable qualification in the discussion hereafter. All formal statements are self-contained and provide exact conditions.

⁵ Note that, together, boundary and interior indivisibilities exhaust the possible forms a non-degenerate indivisibility can take. Any contiguous set of infeasible shares either includes one extreme of the bargaining space or lies in its interior. More general infeasible sets are unions of such indivisibilities, which may coexist in a single game. The impossibility results apply to all such games, as additional infeasibility cannot restore always-peace.

Finally, I show that this problem can be exacerbated when each side’s private information is correlated. Although correlation can introduce a *screening* effect by allowing states to partially infer their opponent’s type, it can also create an *amplifying* effect: when a state receives an extreme signal, it forms more extreme expectations about the stakes and bargains more aggressively. To demonstrate, I explore a special case of a transfer-protocol crisis bargaining game to show how the amplification effect can outweigh the screening effect. Proposition 4 shows the formal result that it can be more difficult to achieve always-peace as information becomes more correlated.

To provide intuition for the main results, I additionally examine a benchmark subclass of crisis bargaining games in which states may not bargain over shares but may agree to a peaceful trade of the object for transfers. This *transfer-protocol* subclass corresponds to crisis bargaining games with the largest possible boundary indivisibility, and I show that always-peaceful equilibria do not exist in this subclass when there is uncertainty about stakes (Proposition 2). These games clarify the incentive to misrepresent private information about the stakes: a payment generous enough to satisfy a state with high expectations invites the same demand from a state with low expectations. Moreover, examining the transfer-protocol subclass reveals that the transfer-receiving state’s private information enters only through its war payoff, so that for it, uncertainty about stakes mirrors uncertainty about power. This analogy is not available when more than one division of the object is feasible in the general class.

1.1 Related literature

The paper makes theoretical contributions to the literature on conflict and political economy. First, I analyze uncertainty about the stakes of the dispute in the crisis bargaining framework with a game-free approach to investigate its implications for every crisis bargaining game. This builds on a tradition that originates with Banks (1990), who characterizes properties of equilibria across all bargaining games when one side has private information about its war payoffs, and later Fey and Ramsay (2011), who study two-sided uncertainty about power and the costs of war, taking the stakes to be fixed and known. Private information about power and costs has dominated the literature on information problems and war since Morrow (1989), and is commonly referred to as the “two standard sources” of uncertainty (Spaniel, 2023). Uncertainty about the dispute’s stakes leads to fundamentally different incentive problems, with new strategic implications.

Second, the results reveal how the structure of dispute resolution processes, including whether shares or transfers are used as instruments, can systematically affect the prospects for peace. I show that the long-standing presumption that transfers can overcome bargaining indivisibilities does not survive the introduction of uncertain stakes. This speaks to the literature in international politics on side payments and issue linkage, from Stein’s (1980) foundational work to later arguments on limits of the effectiveness of linkage strategies (Hassner, 2003; Goddard, 2006).

Third, the extension on correlated information qualifies the standard intuition that reducing uncertainty unambiguously promotes peace (Meirowitz et al., 2022; Sartori, 2002; Ramsay, 2011; Bils, Jordan and Ramsay, 2025) by identifying competing screening and amplification effects of correlated information. I show how always-peace can become more difficult as a result, highlighting that the relationship between information and war is not necessarily straightforward.

Fourth, the paper makes a methodological contribution by presenting a new characterization for the crisis bargaining game form, which allows for an exchange of transfers alongside an allocation of shares in peaceful settlements. The framework also accommodates mixed strategies and stochastic settlements, which previous formulations (most notably, Fey and Ramsay (2011)) do not explicitly admit.⁶ The paper thus advances a growing agenda that employs mechanism design to study features of crisis bargaining settings, including ultimatums (Fey and Kenkel, 2021), intermediate policy responses (Kenkel and Schram, 2025), mediation (Hörner, Morelli and Squintani, 2015; Meirowitz et al., 2019), and multilateral treaties (Morrow and Cope, 2021).

Finally, the paper connects the study of conflict to the economic theory of bilateral trade and partnership dissolution under asymmetric information, where incentive constraints can prevent privately informed parties from realizing gains from trade (Myerson and Satterthwaite, 1983; Chatterjee and Samuelson, 1983; Cramton, Gibbons and Klemperer, 1987). The strategic problem studied here is similar in that private information can prevent states from implementing mutually preferable peaceful settlements even when transfers are unrestricted. The transfer-protocol subclass makes this connection especially direct, while the general class extends the logic to settings where states may also divide the contested object’s value. Crisis bargaining settings differ from those of standard bilateral trade in that war destroys value and the contested object can also be divided rather than

⁶ For their purposes, Fey and Ramsay’s (2011) deterministic mechanisms may be equivalent to stochastic ones, with shares interpreted as averages, but to my knowledge this has not been proven.

only exchanged.

1.2 Information problems and incentives in war

It is well known that a state with uncertainty about an opponent's value for the object faces the same problem as one that faces uncertainty about their opponent's costs of war (Spaniel, 2023, p. 15). Due to this connection between the object's value and the costs of war, it has become standard to normalize the value to 1 for all states and focus on settings where states have uncertainty about each other's costs. My analysis departs from this standard, instead considering uncertainty about the stakes of the dispute, which includes not only uncertainty about an opponent's value for the object but also *their own* value.

To my knowledge, no previous work has studied crisis bargaining when states do not know their own value for the object. More generally, I am unaware of another crisis bargaining study in which a state is uncertain about their own payoff from a peaceful settlement at the time of agreement, in addition to their war payoff and their opponent's payoffs. This is surprising, as this type of uncertainty is ubiquitous in international conflict. Indeed, in any dispute, both sides are unlikely to have exactly the same information about the value of what they are fighting over. Even when, for example, oil reserves on disputed land have been precisely measured and extraction costs are well-understood, the territory's real value depends on qualities of an uncertain future. Profits depend on market prices that reflect global supply and demand, which are in turn dependent on trading relations that evolve alongside geopolitical partnerships. When agreeing to peace or initiating war, states must form expectations about this value, which they do from different informational vantage points. Private information about stakes, like that about power and the costs of war, can generate an incentive to misrepresent that information, opening the door to bargaining failure and war.

A related type of resolve uncertainty, in which states know their own value for the object but do not know *each other's* private valuation, has been explored more widely (Fearon, 1997; Slantchev, 2005). Sartori (2002) studies cheap talk diplomacy in such a setting, and Trager (2011) shows that states can communicate private valuations through what they claim to be willing to fight for. Similarly, Bills and Spaniel (2017) study bargaining under privately known preferences over a spatial policy. Uncertainty about a common value has also been studied with one-sided information, as in Dal Bó and Powell (2009), where a government knows the size of the resources in dispute and offers

a share to an opposition that does not. Accordingly, despite their similarities, these studies do not examine uncertainty about stakes as defined here, where neither side knows the value of the object and hence payoffs that depend on it, whether in war or peace, are unknown to both.

I focus on a jointly unknown *common* value $v_i = v_j = v$ for a state i and their opponent j , which is the case the literature leaves open. I do not cover jointly unknown private values, as this would introduce dual uncertainty about stakes and the resolve uncertainty just described, confounding their effects. Note, any pair of valuations $(v_i, v_j) = (v\beta_i, v\beta_j)$ with costs of war c_i can be rewritten in terms of a common scale component v and a separate resolve component β_i , so that c_i/β_i captures the effective costs of war. Holding the resolve components fixed therefore isolates the effect of uncertainty about stakes. The common value also separates the mechanism from bargaining failures over indivisible objects with private values (Myerson and Satterthwaite, 1983; Cramton, Gibbons and Klemperer, 1987), where surplus depends on which party holds the object and information rents obstruct its efficient allocation. Here, no such allocation problem exists, as every division of the object yields the same surplus.

The common value is especially consequential when we introduce the opportunity to settle with both share allocations and external transfers. Under standard approaches, the terms of a peaceful settlement are denominated in units of the contested object's value by default. When the value is known to be 1, receiving an x share of the object is the same as receiving a transfer worth an x share, which is the same as receiving the full object in exchange for a transfer worth a $1 - x$ share. However, when states do not know the value of the contested object, no such equivalence can be established, as transfers cannot be denominated in units of an unknown quantity.

Uncertainty about the stakes of the dispute is therefore structurally different than these other sources. Table 1 summarizes the distinction. In the standard cases, private information shifts the relative attractiveness of war as an outside option, but not the utility conferred by any given peaceful settlement. In contrast, with uncertainty about stakes, a higher expected value for the object increases both the return to fighting as well as the value for reaching peace at given terms. Both the war payoff and the peace payoff move in tandem with changes to a state's private information at any settlement granting the state a share of the object.

These structural differences produce different incentives. Under the standard sources of uncertainty, states are motivated to overstate their strength (i.e., overstate power and understate costs).

Uncertainty	War payoff	Peace payoff	Terms of peace	Incentive
Power	Unknown, incr. in p_i	Known, constant	Share $\sim$ transfer	Overstate p_i
Costs of war	Privately known, decr. in c_i	Known, constant	Share $\sim$ transfer	Understate c_i
Stakes	Unknown, incr. in v	Unknown, incr. in v	Share $\not\sim$ transfer	Ambiguous

Table 1: Comparison of information problems in crisis bargaining. *Note:* In the payoff columns, the first element indicates whether the payoff is known to all states, privately known, or unknown to all (given terms). The second element indicates how the payoff changes in the uncertain parameter. The terms of peace column indicates whether shares and transfers are commensurable ($\sim$).

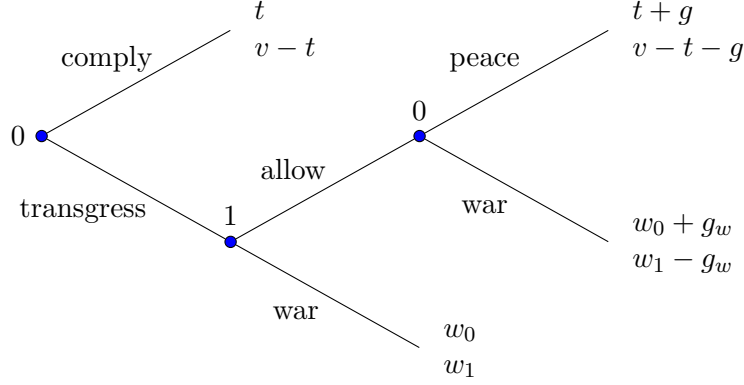
With uncertainty about stakes, however, the incentive to misrepresent has no fixed direction, as which way a state wants to shade its report depends on its bargaining position rather than its type. For example, in a standard ultimatum game over shares, a receiver expecting lower stakes is willing to accept weaker offers, so the proposer wants to understate v and the receiver wants to overstate it.

2 Crisis bargaining games

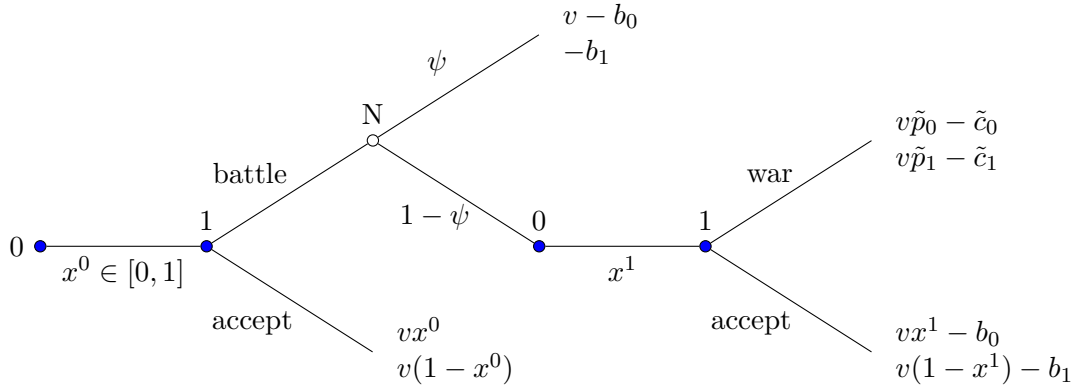
This section presents the class of crisis bargaining games and then introduces uncertainty about stakes. The complete development of the class, including the mechanism design results that support the game-free approach, is presented in Appendix A, with the most relevant items cross-referenced in the footnotes.

A crisis bargaining game Γ is an interaction between two states, $i = 0, 1$, whose actions $a_i \in A_i$ lead to either a peaceful settlement or war. Throughout the paper, I use j to denote a state i 's opponent. Actions and timing are unrestricted, so the class contains ultimatums, alternating offers, wars of attrition, and any other protocol one could imagine. Figure 1 presents examples from the literature to illustrate the breadth of the class across extensive forms.

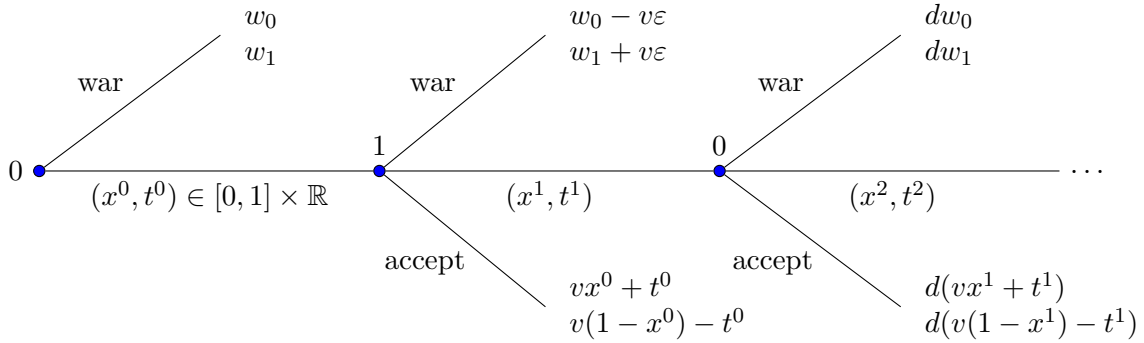
War gives state i an expected payoff of $w_i = v_i p_i - c_i$, where v_i is the value it places on the contested object, p_i is its probability of victory with $p = p_0 = 1 - p_1$, and $c_i > 0$ are the costs of war. A peaceful settlement allocates a share $x_i \in [0, 1]$ of the object and a net transfer $t_i \in \mathbb{R}$ to each state i , yielding payoffs $v_i x_i + t_i$. A share of the object is thus a share of its value, while transfers



(a) Gurantz and Hirsch (2017), with $g, g_w > 0$ as the gains from transgression.



(b) Smith and Spaniel (2019), with $b_i \in (0, c_i)$ as battle costs and ψ as the probability 0 wins decisively after battle. Note that $\tilde{p}_1 = 1 - \tilde{p}_0 = p_1/(1 - \psi)$ and $\tilde{c}_i = (c_i - \psi b_i)/(1 - \psi)$ establishes that state i 's expected payoff from battle given subsequent war is w_i .



(c) Powell (2006), with $d \in [0, 1]$ as the discount factor and $2\epsilon \in (0, 2\min_i p_i)$ as the first-strike advantage.

Figure 1: Examples of crisis bargaining games. *Note:* The game trees depict games that are similar in spirit to those in the corresponding articles, but the specifications have been slightly altered from the original versions to establish clear connections with the present analysis (e.g., Powell (2006) did not originally feature transfers). The superscripts on proposals index periods.

are external to the object and hence have a value independent of the object's. As usual, peace is efficient, so shares must sum to one and transfers must net to zero, $x_0 = 1 - x_1$ and $t_0 = -t_1$.

An outcome of the game is therefore a triple (π, x, t) , where $\pi \in \{0, 1\}$ indicates whether war occurs and $(x, t) = (x_0, t_0)$ describes the settlement in peace. Each profile of actions $a = (a_0, a_1) \in A$ induces a distribution $\gamma(a)$ over these outcomes. Each state privately observes a type $\theta_i \in \Theta_i$, with the type pair θ drawn from a common joint prior F , and any of the primitives may depend on the types. When the dependence matters, the primitives are written as functions of the type profile, such as $w_i(\theta)$ for the war payoff under θ . After presenting a benchmark result, I introduce uncertainty about stakes, the case of interest.

The only substantive assumption on the class is that agreements are voluntary, i.e., no state can be compelled to accept terms it expects to be worse than war.⁷ Formally, each state i always has some action $\tilde{a}_i$ available such that, for every type θ_i ,

$$\mathbb{E}_{\theta_j}[u_i((\tilde{a}_i, a_j(\theta_j)), \theta) \mid \theta_i] \geq \mathbb{E}_{\theta_j}[w_i(\theta) \mid \theta_i] \quad (\text{VA})$$

given any opposing play $a_j : \Theta_j \rightarrow A_j$, where $u_i(a, \theta)$ denotes state i 's expected payoff under the profile a at type pair θ . Most commonly, states have the option of starting a war. The VA criterion here is weaker than usual in that it does not require a state's realized peace payoff to exceed their war payoff, but only that states have actions available that, if taken, would guarantee at least their war payoff in expectation. In particular, a state may receive less in peace than what they would have demanded *had they known* the truth. I return to a stronger notion in the section on interior indivisibilities to examine whether the settlements that can always be reached can also be maintained.

The analysis proceeds through direct mechanisms, in which states simply report their private types and a mapping from reported types to distributions over outcomes determines war, shares, and transfers. For a mechanism δ , the induced probability of war is given by $\pi^\delta(\theta) \in [0, 1]$ and, in a peaceful settlement, $x_i^\delta(\theta) \in [0, 1]$ and $t_i^\delta(\theta) \in \mathbb{R}$ give state i 's expected share of the object and

⁷ Assumption A.1 states the assumption explicitly.

expected net transfer, respectively. Given these, we can denote by

$$u_i^\delta(\tilde{\theta}_i; (\theta_i, \theta_j)) = \underbrace{\pi^\delta(\tilde{\theta}_i, \theta_j)}_{\text{prob war}} \underbrace{w_i(\theta_i, \theta_j)}_{\text{war payoff}} + \underbrace{(1 - \pi^\delta(\tilde{\theta}_i, \theta_j))}_{\text{prob peace}} \underbrace{(v_i(\theta_i, \theta_j) x_i^\delta(\tilde{\theta}_i, \theta_j) + t_i^\delta(\tilde{\theta}_i, \theta_j))}_{\substack{\text{value of share} \\ \text{peace payoff}}} \quad (1)$$

the expected utility of a state with type θ_i reporting as $\tilde{\theta}_i$ at a realized type pair θ . From this, their interim expected utility is defined as⁸

$$U_i^\delta(\tilde{\theta}_i; \theta_i) = \int_{\Theta_j} u_i^\delta(\tilde{\theta}_i; (\theta_i, z)) dF(z | \theta_i). \quad (2)$$

A mechanism is incentive compatible if every type is willing to truthfully report their type.⁹ By the revelation principle (Myerson, 1979) and a result establishing its converse (Lemma S.1, SI 1.1), there exists a crisis bargaining game that admits an *always-peaceful equilibrium*, in which war never occurs on the path of play, if and only if there exists an incentive-compatible mechanism that never produces war and leaves every type of each state weakly preferring its expected settlement to what it expects in war.¹⁰ Direct mechanisms are hereafter understood to satisfy this interim participation constraint. Exhibiting such a mechanism thus establishes the existence of a game and equilibrium, while showing that no such mechanism exists establishes impossibility for the entire class.

Now that the class is established, let us briefly establish that indivisibilities in the object, as defined by this paper, do not affect the prospects of peace under standard sources of uncertainty. When both states place the same value v on the contested object, I refer to this common value as the *stakes* of the dispute. When the stakes are known, transfers can substitute for any infeasible division at a known rate, regardless of any other uncertainty states face. The result is stated as follows,¹¹ with all proofs and auxiliary results presented in the Supplemental Information.

Proposition 1. *When the stakes of the dispute v are known, no restriction on the feasible divisions of the object changes the probabilities of war or the payoffs attainable across crisis bargaining games, even as uncertainty about power or the costs of war persists.*

⁸ Equations (A.1) and (A.2) specify the game-form counterparts $u_i(\cdot)$ and, given strategies σ , $U_i^\sigma(\cdot)$.

⁹ Definition A.1 defines incentive compatibility.

¹⁰ Definition A.2 defines an always-peaceful equilibrium, and Remark A.2 provides the statement that connects the existence of an incentive-compatible direct mechanism to the existence of a crisis bargaining game that admits an always-peaceful equilibrium.

¹¹ Proposition B.1 in Appendix B presents a more formal statement of Proposition 1.

Any probability of war and any payoffs achievable in a crisis bargaining game remain achievable when any set of divisions is infeasible. The standard sources of information problems in crisis bargaining may still cause war, but restrictions on divisions contribute nothing to the problem, as unconstrained transfers substitute perfectly for the object at its known value. The rest of the paper studies the case in which the stakes are unknown, and no such substitution is available.

2.1 Games with uncertainty about stakes

We can now turn to games with uncertainty about stakes. Let the probability of victory p and the costs of war c_i be common knowledge. On the other hand, both states have private information about the common value of a contested object, given by a Borel measurable function $v : \Theta \rightarrow \mathbb{R}$. I assume that v is continuously differentiable and weakly increasing in both of its arguments, and that the object is always worth enough to justify fighting over it, i.e., $\min_{\theta} v(\theta) \geq \max_i c_i/p_i$. This latter condition is sufficient to avoid uninteresting cases where peace prevails due to expectations that the object is not valuable enough to dispute over, falling outside of the purview of crisis bargaining.

Recall that a type pair θ is drawn from a common joint prior F on Θ . For a state i with type θ_i , the interim expected stakes are given by

$$V_i(\theta_i) = \int_{\Theta_j} v(\theta_i, z) dF(z | \theta_i).$$

Letting $\bar{\theta}_i \equiv \max \Theta_i$ and $\underline{\theta}_i \equiv \min \Theta_i$ for each state i , we can then express the interim expected stakes of the highest and lowest types by $\bar{V}_i = V_i(\bar{\theta}_i)$ and $\underline{V}_i = V_i(\underline{\theta}_i)$, respectively.

Of course, additional information about an opponent's type can usually be gleaned through the sequence of play. For example, a settlement offer can itself reveal information about the opponent, and these posterior beliefs shape the decision to cooperate or fight. Different crisis bargaining games will therefore present different opportunities to learn information. Regardless of what states learn throughout a particular game, voluntary agreements require an always-peaceful equilibrium to leave every type weakly preferring its expected settlement to its interim expected war payoff.

For this reason, we do not need to deal directly with posterior beliefs that emerge from specific games. An incentive-compatible direct mechanism that never produces war and satisfies participation at every posterior belief realized on the path of play must also satisfy the necessary interim

participation constraint $U_i^\delta(\theta_i; \theta_i) \geq V_i(\theta_i)p_i - c_i$ for each state i and all $\theta_i \in \Theta_i$ (Lemma S.2, SI 1.2). Thus, to consider the existence of always-peaceful equilibria, we can simply consider this necessary condition for always-peace that depends exclusively on each state's interim expected utilities. If there does not exist an incentive-compatible direct mechanism that satisfies the interim participation constraint for all types of each state, there does not exist an always-peaceful equilibrium to any crisis bargaining game with uncertainty about stakes.

Remark 1. *When there is uncertainty about stakes and every division of the object is feasible, there exists a crisis bargaining game that admits an always-peaceful equilibrium.*

With no indivisibility, there is a crisis bargaining game in which states settle at shares of the object matching the known balance of power. A settlement of $x = p$ with no transfers gives each state i a peace payoff of $V_i(\theta_i)p_i$, which weakly exceeds their expected war payoff for every type, and no state can gain by misrepresenting their information under a constant settlement. Uncertainty about stakes without an indivisibility therefore does not preclude peace. Notably, this settlement survives even when transfers are banned outright, in which case a state's private information matters only through its costs of war relative to the object's value, and uncertainty about stakes operates like uncertainty about the costs of war (see Proposition S.1 in the Supplemental Information).

The formulation is also deliberately permissive for achieving always-peace. In addition to the weaker form of the VA criterion, I allow for stochastic mechanisms, in which the final settlement is drawn from a distribution over outcomes, so that peaceful agreements can be reached by averaging across settlements not acceptable on their own. The impossibility results are therefore established in their strongest form, holding under conditions most favorable to peace.

3 Boundary indivisibilities

Now we can examine whether transfers can facilitate peaceful agreement when share allocations sufficient to guarantee a mutually acceptable settlement are infeasible. To explore these questions, I focus on de facto indivisibilities in which some share allocations cannot be implemented. I remain agnostic about their source, be it a literal indivisibility, an institutional constraint, or another feature of the strategic interaction. All that matters is that some share allocations are infeasible. It is worth reiterating that these are not indivisibilities in the usual payoff-based sense. Unconstrained transfers

still allow states to implement any division of the *surplus* from peace given the shares that remain, so no peace payoffs are ever placed out of reach.

These arise in two possible ways: boundary and interior indivisibilities. Let us first focus on any crisis bargaining game with a boundary indivisibility. These are cases in which extreme share allocations to one side are not feasible. For example, this may arise in situations where one state cannot peacefully concede too much of the object due to its importance for their sovereignty and continued survival. Figure 2 presents examples.

Definition 1 (Boundary Indivisibility). *A crisis bargaining game Γ has a boundary indivisibility if and only if there exists an $\bar{x} \in (0, 1]$ such that, for a given state i and every $a \in A$, $x_i \geq \bar{x}$ for all $(0, x, t) \in \text{supp}(\gamma(a))$.*

In principle, a dispute could have two boundary indivisibilities, but we can focus on the case of one without loss of generality, as additional infeasibilities cannot recover always-peace. Note that, to reduce the notational burden, I adopt the convention that state i is the side unable to peacefully concede extreme shares of the object, so that $\bar{x}$ is the minimum share that can be peacefully allocated to i , without loss of generality. Moreover, every crisis bargaining game with a boundary indivisibility such that $\bar{x} = 1$ is in the transfer-protocol subclass.

The tension created by a boundary indivisibility is straightforward: to achieve always-peace whenever such an indivisibility exists with corresponding threshold $\bar{x}$ for some state i , then the opponent j must always be willing to accept a settlement in which they are allocated a share of no more than $1 - \bar{x}$. But some types of the opponent may expect enough from the object that shares below $1 - \bar{x}$ leave them preferring war unless transfers make up the difference. Can transfers help facilitate peace?

To evaluate this question for boundary indivisibilities, I proceed with an assumption that private types are drawn independently. This assumption is common in previous work, most notably in Fey and Ramsay (2011), and is helpful to ensure there is sufficient uncertainty about the stakes of the dispute. If types are highly correlated, states may be able to precisely infer their opponent's type, and hence the object's value, from their own information. In a later section, I show that correlated information can make always-peace even more difficult to achieve, as long as the screening effect just described is sufficiently weak.

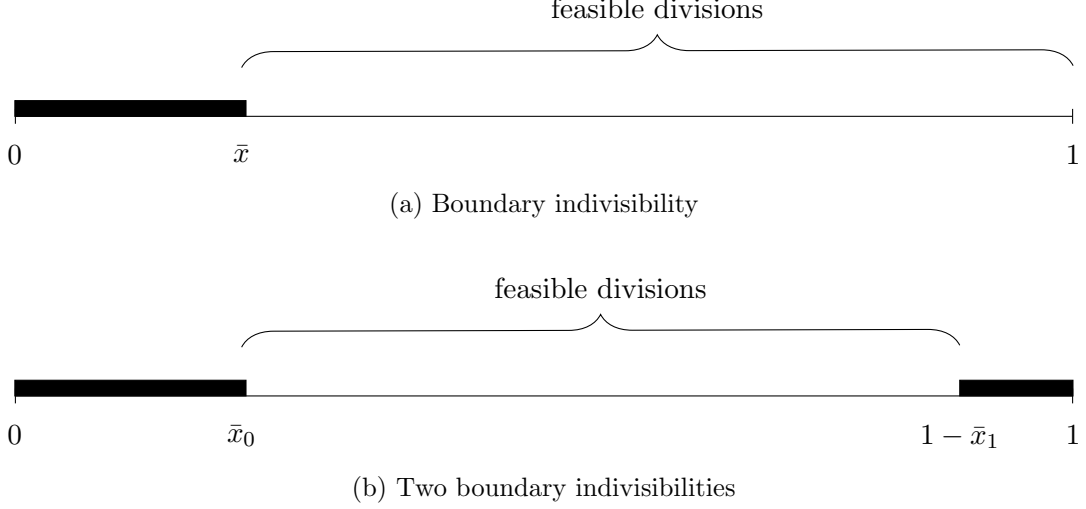


Figure 2: Examples of boundary indivisibilities in the space of divisions $x_0 \in [0, 1]$. *Note:* There is a boundary indivisibility when, for all $a \in A$, every realized outcome with $\pi = 0$ in the support of $\gamma(a)$ leaves a state i with at least $\bar{x}_i$ of the object. In panel (a), state 0 must receive at least $\bar{x}$, and in panel (b), each state i must receive at least $\bar{x}_i$, so feasible divisions give state 0 a share in $[\bar{x}_0, 1 - \bar{x}_1]$.

Assumption 1 (Independent Types). $F(\theta) = F_0(\theta_0)F_1(\theta_1)$.

3.1 Boundary case: Peaceful agreements cannot always be reached

For a transfer scheme to be implementable within an equivalent direct mechanism, it needs to satisfy both the most pessimistic type of state i that expects the lowest stakes and the most optimistic type of opponent j that expects the highest stakes. Transfers can vary type pair to type pair, but must maintain incentive compatibility: as the transfer received by one type of opponent j becomes much larger than that received by another, the types receiving less will mimic the types receiving more unless these incentives are balanced by their expected shares. If these incentives cannot be balanced using shares and transfers, there will be a risk of war in equilibrium. The following result characterizes sufficient interim conditions that rule out the possibility of an always-peaceful equilibrium in every crisis bargaining game.

Theorem 1. *No crisis bargaining game with uncertain stakes and a boundary indivisibility admits an always-peaceful equilibrium unless the costs of war are sufficiently large. Formally, suppose power p and the costs of war (c_0, c_1) are common knowledge, but there are uncertain stakes v . If Assumption*

1 holds and there is a boundary indivisibility with $\bar{x} \in (0, 1]$ such that

$$c_0 + c_1 < (\bar{x} - p_i)(\bar{V}_j - \underline{V}_i) - (1 - \bar{x})(Ev - \underline{V}_j) \quad (3)$$

for a state i and opponent j , then there does not exist an incentive-compatible direct mechanism δ such that, for all $\theta \in \Theta$, $\pi = 0$ and $x_i \geq \bar{x}$ for all $(\pi, x, t) \in \text{supp}(\delta(\theta))$.

Regardless of what conflict resolution procedure is in place, private information about the stakes of the dispute induces states to trade off a risk of war for an opportunity to receive more lucrative settlements in peace. It is not that always-peace is impossible in principle, as mutually preferable settlements must exist for any realized stakes, but there are no incentive-compatible mechanisms that can be instituted to consistently deliver them. Notably, always-peace here is defined in terms of reaching agreements and not maintaining them, and so always-peace is precluded even when states can commit to lotteries over settlements and to maintaining unfavorable realized terms.

Inequality (3) provides a mild condition that prevents always-peace with a boundary indivisibility, requiring in essence that the costs of war not be too large relative to the indivisibility. It is unsurprising that we require a condition of this kind, as any peaceful settlement is trivially preferred to war when the costs of war are made arbitrarily large, stripping the setting of its strategic content.

To unpack the condition, consider the right-hand side of inequality (3). The first component corresponds to the compensation problem states face in attempting to resolve the dispute by dividing the pie. The quantity $\bar{x} - p_i$ reflects the minimum distance a settlement's share must be from the "power-balanced" share $x = p$, and it interacts with $\bar{V}_j - \underline{V}_i$, the difference between the highest possible expected stakes of one state and the lowest possible expected stakes of another. Together, the first term captures the size of the problem created by the indivisibility. Transfers can, in principle, be used to make the opponent j whole and overcome this imbalance in shares, but as in the transfer-protocol subclass, they invite mimicry unless the types receiving larger transfers also receive a sufficiently smaller share. The term $1 - \bar{x}$ gives the maximum share that can be allocated to the short-changed side, governing the flexibility any mechanism designer has to vary j 's shares to maintain incentive compatibility. This interacts directly with $Ev - \underline{V}_j$, which governs how much additional value j attributes to the shares on average.

An implication of Theorem 1 is that even small indivisibilities in the object can be an insur-

mountable obstacle to always-peace when there is a high power differential between the states. In particular, when the weaker state is arbitrarily weak, always-peace can be impossible even as the indivisibility becomes arbitrarily small. Whether or not such a situation is realized depends on the relationship between private information and the stakes of the dispute: it is most difficult to achieve always-peace when state i 's lowest type expects very small stakes, the opponent j 's highest type expects very large stakes, and j 's lowest type expects stakes to be close to the ex-ante expectation.

One intuitive way for this to occur is if the stronger state's private information can unilaterally swing the stakes upward far beyond the ex-ante expectation. To see this, consider a simple example in which $v(\theta_i, \theta_j) = \theta_i + \theta_j^2$, where θ_i is distributed uniformly on $[1, 3]$ and θ_j is drawn to be 0 with probability 999/1000 and 20 with probability 1/1000 (or any continuous approximation). Here, the object in dispute is a jackpot with very small probability, and opponent j alone has private knowledge of whether it is. Simple calculations reveal that the lowest expectation state i can have, $7/5$, is significantly below the highest expectation state j can have, 402. Further, the ex-ante expectation is $12/5$, which is close to 2, the expected stakes of j 's lowest type. Even if the indivisibility hardly blocks the power-balanced division, the right-hand side of inequality (3) can be large.

To pin down a specific case, take $p_i = 1/200$ and $\bar{x} = 1/100$.

Then, the right-hand side of (3) becomes approximately $8/5$. Even though the indivisibility is very small in absolute terms, only preventing settlements that allocate more than 99% of the object to opponent j , an always-peaceful equilibrium is not possible unless the total costs of war are larger than roughly two-thirds of the ex-ante expected value of the object. Indeed, under the maintained lower bound on the value of the object to avoid trivial settings, every admissible cost pair lies below this threshold. Increasing the value of the unlikely jackpot further worsens the prospect of always-peace, requiring even greater total costs of war.

3.2 Benchmark: a complete indivisibility

Theorem 1 establishes the general impossibility result for crisis bargaining games with boundary indivisibilities. To isolate the incentive problem behind this result, I now study the limiting case of a boundary indivisibility under uncertainty about stakes. Specifically, I consider the benchmark subclass of crisis bargaining games in which settlements may only specify a transfer in exchange for the object, which I call the *transfer-protocol* subclass.

$\theta_0 \sim F_0$		
$\theta_1 \sim F_1$	(θ_0, θ_1)	fight settle
	fight	$w_0(\theta), w_1(\theta)$ $w_0(\theta), w_1(\theta)$
	settle	$w_0(\theta), w_1(\theta)$ $v(\theta) - t, t$

Figure 3: Example of a transfer-protocol crisis bargaining game with uncertain stakes. State 0 (row) and 1 (column) simultaneously choose to fight or settle. Dashed boxes reflect state uncertainty about their opponent's type. Fighting yields war payoffs, whereas settling returns state 0 with the object in exchange for a fixed transfer t .

This is in effect the largest possible boundary indivisibility in which $\bar{x} = 1$, as a designated state must receive the entire object and no negotiation over its division is possible. When the value of the object is known, transfer protocols are indistinguishable from other protocols that allow for division of the object: receiving x share of an object of value v is equivalent to receiving the object and extending a transfer equal to $v(1 - x)$. However, as this section demonstrates, the distinction has important consequences when states are uncertain about the value of the object.

Definition 2 (Transfer Protocol). *A crisis bargaining game Γ uses a transfer protocol if and only if, for every $a \in A$ and $(\pi, x, t) \in \text{supp}(\gamma(a))$, $x_i = 1$ for a given state i whenever $\pi = 0$.*

Let us now consider a simple example of a transfer-protocol crisis bargaining game, shown in Figure 3. States observe their private type and simultaneously choose whether to fight or settle. If at least one chooses to fight, each state receives their war payoff $w_i(\theta)$. On the other hand, if both states choose to settle, state 0 receives the entire contested object of value $v(\theta)$ in exchange for a fixed transfer t to state 1.

Given their opponent is settling, state 1 settles if and only if the transfer compensates them for their expected war payoff, $t \geq E_{\theta_0} w_1(\theta)$. On the other hand, state 0 settles only if the transfer they extend is smaller than the additional value they expect to receive from peacefully acquiring the object, $t \leq E_{\theta_1} [v(\theta) - w_0(\theta)]$. A single fixed transfer must hold for all types, satisfying both the highest type of the receiving state and the lowest type of the extending state. This is impossible when the expected war payoff of state 1's high type is larger than their opponent's expected gain from settling peacefully, i.e., $c_0 + c_1 < p_1(\bar{V}_1 - \underline{V}_0)$.

This condition parallels a sufficient condition for the impossibility of always-peace in any crisis

bargaining game that involves trading the object to state 0 for a transfer. More generally, we can state the following result.

Proposition 2. *No transfer-protocol game with uncertain stakes admits an always-peaceful equilibrium unless the costs of war are sufficiently large. Formally, suppose power p and the costs of war (c_0, c_1) are common knowledge, but there are uncertain stakes v . If Assumption 1 holds and*

$$c_0 + c_1 < p_j(\bar{V}_j - V_i)$$

for a state i and opponent j , then there does not exist an incentive-compatible direct mechanism δ such that, for all $\theta \in \Theta$, $\pi = 0$ and $x_i = 1$ for all $(\pi, x, t) \in \text{supp}(\delta(\theta))$.

With transfer protocols, the extending state i , in walking away with the object, captures the upside of more favorable private information directly. In contrast, opponent j , in walking away with a transfer, cannot. A transfer scheme that tries to compensate higher types of j with higher transfers would invite mimicry from low types, who would strategically exaggerate their demands to capture larger transfers, as well. Anticipating this, state i would be less willing to extend large transfers in the first place. The receiving state's transfer must therefore be constant in their private information. Corollary S.2 (SI 1.4) shows that the only way for the receiving state's interim utility to vary in their type is then through the probability of war, but this channel would also be shut down in any always-peaceful equilibrium.

The condition in Proposition 2 is a direct consequence of this fact. Once expected transfers are forced to be constant across the receiving state's types, the question of whether always-peace is possible reduces to whether a single transfer can simultaneously satisfy the most demanding type of the receiving state and the least willing type of the extending state. With enough variation in the expected stakes across different types relative to the costs of war, no constant transfer can satisfy all types of both states, and always-peace is impossible.

This is not a statement that no peaceful settlement exists in principle. On the contrary, one must exist: a transfer of $t_j = vp_j$ is guaranteed to satisfy all parties. The problem is that states do not know the stakes, and the asymmetric bargaining positions preclude any transfer scheme that can overcome each side's incentive to misrepresent their private information, leading to war with positive probability in every equilibrium of every possible game.

Note that, in transfer-protocol games, the transfer-receiving state's private information enters only through its war payoff, so it faces uncertainty of the same kind as uncertainty about power, in which peace payoffs are known at the time of agreement. However, the state recovering the contested object faces a deeper uncertainty, as the stakes matter to it in both war and peace. This structure is particular to transfer protocols and does not extend to the general class, where settlements may allocate shares of the object, so the unknown value also enters the peace payoffs.

4 Interior indivisibilities

Now that we have established the impossibility result for boundary indivisibilities and have illustrated the mechanism in the transfer-protocol subclass, let us consider interior indivisibilities, in which extreme settlements can remain possible in each direction. In this case, it is possible for states to agree to peaceful settlements that allocate extreme shares of the object to either party. Figure 4 presents examples.

Definition 3 (Interior Indivisibility). *A crisis bargaining game Γ has an interior indivisibility if and only if there exist $\underline{x}$ and $\bar{x}$ with $1 > \bar{x} > \underline{x} > 0$ such that, for every $a \in A$ and all $(0, x, t) \in \text{supp}(\gamma(a))$, $x \notin (\underline{x}, \bar{x})$.*

As before, the restriction concerns only the divisions of the object that can be implemented, with transfers and hence allocations of the surplus from peace remaining unrestricted.

4.1 Interior case: Peaceful agreements can always be reached

Unlike its boundary counterpart, interior indivisibilities that preclude settlements at the power-balanced share do not inherently bias the allocation of shares toward one side. As a result, conflict resolution mechanisms have more flexibility in reaching agreement. Because the mechanism designer can allocate a more-than-power-balanced share to either side, it has an additional degree of freedom in varying shares across type profiles so that every type of each state expects a favorable share at least some of the time. Because peace is efficient, the mechanism designer can strategically design a settlement scheme so that the gains and losses across these different realizations of type profiles wash out on average, leaving each type of each state expecting more from peace than their war payoff.

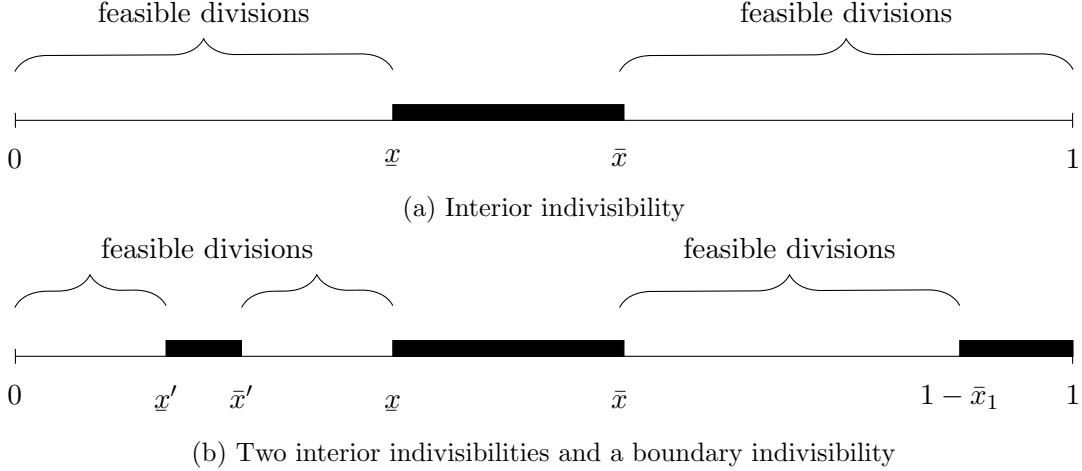


Figure 4: Examples of interior indivisibilities in the space of divisions $x_0 \in [0, 1]$. *Note:* There is an interior indivisibility when, for all $a \in A$, no realized outcome with $\pi = 0$ in the support of $\gamma(a)$ involves a share between $\underline{x}$ and $\bar{x}$. In panel (a), state 0 must receive a share in $[0, \underline{x}] \cup [\bar{x}, 1]$. Panel (b) shows a game whose infeasible set, $(\underline{x}', \bar{x}') \cup (\underline{x}, \bar{x}) \cup (1 - \bar{x}_1, 1]$, is a union of several indivisibilities. The impossibility results apply whenever the conditions are satisfied by any one indivisibility.

To see why agreements can always be reached in this case, consider a simple example. Suppose each side wins a war half of the time, but that shares that allocate more than 40% but less than 60% of the object to either side are infeasible; that is, $p = 1/2$ and $(\underline{x}, \bar{x}) = (2/5, 3/5)$. Then, even though the mechanism designer cannot evenly divide the object for any particular type profile realization, they can still implement a settlement scheme so that each type of each state expects an equivalent payoff from peace over all possible realizations. One such way the mechanism designer could do this is by simply flipping a coin and allocating 60% to the winner of the coin flip. The following proposition formally states the result.

Proposition 3. *For any interior indivisibility, there exists a crisis bargaining game with uncertain stakes that admits an always-peaceful equilibrium. Formally, suppose power p and the costs of war (c_0, c_1) are common knowledge, but there is uncertainty about stakes v . If there is an interior indivisibility given by any $\underline{x} \in (0, 1)$ and $\bar{x} \in (\underline{x}, 1)$, then there exists an incentive-compatible direct mechanism δ such that $\pi = 0$ and $x \notin (\underline{x}, \bar{x})$ for all $(\pi, x, t) \in \text{supp}(\delta(\theta))$, for all $\theta \in \Theta$.*

To overcome an interior indivisibility, the proof uses an example in which the mechanism designer implements a randomization device over the final settlement, providing extreme shares and corresponding transfers with probabilities that result in an expected payoff commensurate with a division that reflects the balance of power. This “randomization-device” mechanism, a variation of

the ones proposed by Fearon (1995), Powell (2006), and Wagner (2010), is consistent with the crisis bargaining game form only because our definition is deliberately weak and does not rule out credible commitments. More generally, the VA criterion requires states only to expect peace to be better than war on average and therefore permits mechanisms in which some types would have to honor realized settlement terms they know to be worse than war. Crisis bargaining usually takes place in an anarchic setting and rarely features this kind of commitment power, as there is no enforcement power that can require states to keep to unfavorable realizations. When settlement terms are revealed that leave a state worse off than war, it can abandon the agreement and fight.

4.2 Interior case: Peaceful agreements cannot always be maintained

Then, an interesting question to explore regarding interior indivisibilities is not whether states can agree to a peace they expect to be better than war on average, but whether any such peace can be maintained once the terms of settlement are realized. The following notion of robustness withdraws the commitment power granted by the baseline framework, requiring settlements that states would both reach *and maintain*. When incorporating this notion of maintaining settlements after an initial agreement, always-peace becomes impossible under interior indivisibilities, as well.

Definition 4 (Robustness). *A direct mechanism δ is called robustly implementable if and only if, for each state i , it satisfies:*

1. *No randomization over the settlement: for all $\theta \in \Theta$, there exists a $(\pi^\delta(\theta), x^\delta(\theta), t^\delta(\theta)) \in \Omega$ such that $\text{supp}(\delta(\theta)) = \{(\pi^\delta(\theta), x^\delta(\theta), t^\delta(\theta))\}$.*
2. *Ex-post incentive compatibility: for all $\theta_i, \tilde{\theta}_i \in \Theta_i$, $\theta_j \in \Theta_j$,*

$$u_i^\delta(\theta_i; (\theta_i, \theta_j)) \geq u_i^\delta(\tilde{\theta}_i; (\theta_i, \theta_j)). \quad (\text{EPIC})$$

3. *Ex-post voluntary agreements: for all $\theta_i \in \Theta_i$, $\theta_j \in \Theta_j$,*

$$u_i^\delta(\theta_i; (\theta_i, \theta_j)) \geq w_i(\theta_i, \theta_j). \quad (\text{EPVA})$$

An equilibrium σ^ of a crisis bargaining game is called robust if and only if the outcome it induces admits a robustly implementable equivalent direct mechanism.*

Although the three conditions of robustness strengthen the benchmark, they largely return the analysis to the defaults of the crisis bargaining literature. The first requires a deterministic settlement: states must know the exact share and transfer they are agreeing to, ruling out any commitment to honor the realization of a lottery over multiple peace deals. This is the default of previous game-free analyses of crisis bargaining (Fey and Ramsay, 2011; Fey and Kenkel, 2021; Kenkel and Schram, 2025).

Second, ex-post incentive compatibility (EPIC) strengthens our notion of incentive compatibility by requiring each state to prefer truthful reporting *given* the realized stakes rather than merely in expectation. In studying “robust” mechanism design that develops results free of any particular specification of beliefs, Bergemann and Morris (2005) show that this stronger ex-post condition delivers incentive compatibility under every possible information structure governing beliefs about each other’s types.

Third, ex-post voluntary agreements (EPVA) likewise strengthen the baseline VA criterion from an interim to an ex-post condition. Under VA, a state accepts a settlement it expects to serve it, though the realization may prove worse than war; under EPVA, no realization does, so a settlement acceptable when reached remains acceptable once the terms are known. This is the sense in which the robustness conditions require agreements to be maintained and not merely reached.

Under robustness, always-peace must be a stable outcome not just on average, but at every realization the mechanism may produce. The following result establishes that, under these robustness conditions, interior indivisibilities preclude always-peaceful equilibria in every crisis bargaining game with uncertain stakes.

Theorem 2. *No crisis bargaining game with uncertain stakes and an interior indivisibility admits a robust always-peaceful equilibrium unless the costs of war are sufficiently large. Formally, suppose power p and the costs of war (c_0, c_1) are common knowledge, but there are uncertain stakes v . Further, let v be strictly increasing in both of its arguments and Θ_i be a compact interval for each state i . Then, if there is an interior indivisibility given by $\underline{x} \in (0, 1)$ and $\bar{x} \in (\underline{x}, 1)$ such that*

$$c_0 + c_1 < \frac{(\bar{x} - p)(p - \underline{x})}{\bar{x} - \underline{x}} \max_i \left\{ \max_{\theta_j} [v(\bar{\theta}_i, \theta_j) - v(\underline{\theta}_i, \theta_j)] \right\}, \quad (4)$$

there does not exist a robustly implementable direct mechanism δ such that $\pi = 0$ and $x \notin (\underline{x}, \bar{x})$ for

all $(\pi, x, t) \in \text{supp}(\delta(\theta))$, for all $\theta \in \Theta$.

Under the robustness conditions, the mechanism designer no longer has the lottery mechanism available to them, and they must choose a peaceful settlement scheme so that shares never fall within the indivisibility, transfers compensate states that receive unfavorable shares, and states have the incentive to truthfully report their type. In this setting, EPIC forces share allocations in which higher types receive more favorable shares. The intuition is that a share is worth more to a state expecting higher stakes, while a transfer is worth the same to every type. A settlement carrying a more favorable share is therefore disproportionately attractive to higher types, and no fixed transfer can make up for a less favorable share across all of them, so truthful reporting requires the more favorable shares to go to the higher types. Due to the interior indivisibility, share allocations must then “jump” from weakly below $\underline{x}$ to weakly above $\bar{x}$ as state 0’s type increases past the threshold, and vice versa for state 1.

Consider types just above and below the threshold. Both must prefer their assigned settlement to the other’s and, because the higher type values the object more, incentive compatibility requires the difference between what each receives in peace to grow as the stakes of the dispute rise. But the EPVA criterion puts a constraint on that difference. Because every type of every state must always prefer peace to war, the difference in peace payoffs between any two types can exceed the difference in their war payoffs by no more than the total costs of war. When there is sufficient variation in stakes across types, EPIC demands more variation in states’ peace payoffs relative to war than EPVA allows.

Inequality (4) reveals that impossibility may result even with arbitrarily small indivisibilities around the power-balanced share, as long as there is sufficient variation in the object’s value. It is also worth noting that, unlike the preceding impossibility results, Theorem 2 does not make any assumptions about the distribution of private information; the criteria for a robust always-peaceful equilibrium cannot be jointly satisfied under any joint distribution F .

Interior indivisibilities are therefore more forgiving than boundary ones, but only by virtue of commitment power that is usually assumed not to exist in international interactions. Once settlements must survive the realization of their own terms, neither form of indivisibility permits always-peaceful equilibria.

5 Correlated information

Now, let us focus on the effect of correlated signals. I identify two countervailing forces: one in *screening*, in which more information about the opponent's type helps facilitate peaceful agreement, and another through *amplification*, in which states with extreme signals intensify their demands, making it more difficult to successfully reach an agreement and avoid war.

With additional regularity conditions (e.g., supermodularity of v), one could derive general impossibility results comparable to Theorems 1 and 2, but I instead focus on a special case, which suffices to demonstrate that the less-intuitive amplification effect can outweigh the more-intuitive screening effect and exacerbate the problem. The case is stark, as incentive compatibility eliminates the screening effect while the amplification effect persists (Lemma S.7). To begin, let us specify the distribution of private information.

Assumption 2 (Correlated Types). *Types θ_0, θ_1 are drawn from the joint density*

$$f(\theta_0, \theta_1; \lambda) = \frac{1}{4}(1 + \lambda\theta_0\theta_1)$$

on Θ , with $\Theta_0 = \Theta_1 = [-1, 1]$ and $\lambda \in [0, 1]$.

By this assumption, the degree of correlation is measured by a dependence parameter λ , with $\lambda = 0$ corresponding to independent types and larger λ implying a higher correlation between each state's type; specifically, $\text{corr}(\theta_0, \theta_1) = \lambda/3$. While this specification focuses on positively correlated information, similar results can be attained with negatively correlated information, as well. This specification is useful because the marginal distribution of each state's own type is invariant to λ , so changing λ changes only the linkage between signals, not the distribution each state's type is drawn from. Because the stakes depend on both signals, changing their correlation can also change the ex-ante distribution of the object's value. Concretely, if states privately know how much gold is on their side of a territorial dispute, increases in λ make gold on one side more predictive of gold on the other side without changing the amount of gold expected on either side ex ante. Figure 5 presents examples of the implied conditional cumulative distribution.

Next, let us further specify how private information relates to the dispute's stakes.

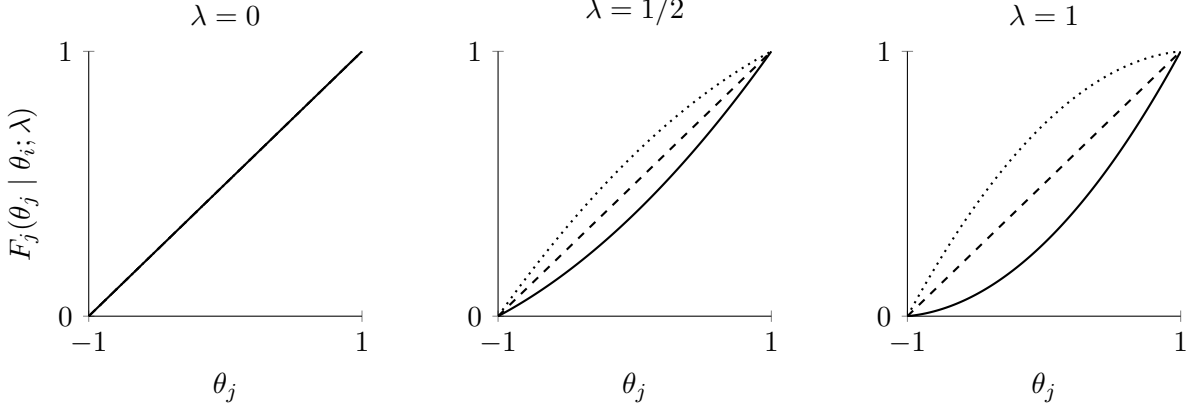


Figure 5: Examples of the conditional CDF $F_j(\theta_j | \theta_i; \lambda)$ implied by Assumption 2. *Note:* From left to right, the plots show distributions under zero, moderate, and high correlation, reflected by changes in the dependence parameter λ . The dotted, dashed, and solid lines reflect the distribution of θ_j given a low, moderate, and high value of θ_i , respectively.

Assumption 3. $v(\theta_0, \theta_1) = \phi(\theta_0 + \theta_1)$ where $\phi : [-2, 2] \rightarrow \mathbb{R}$ is strictly increasing with $\phi(-2) > \max_i c_i/p_i$.

Letting the value of the contested object depend on the sum $\theta_0 + \theta_1$ is a simplifying assumption that symmetrically reduces the type pair to a single scalar. The maintained monotonicity is strengthened to hold strictly so that favorable information always raises the stakes. Moreover, the existing lower bound on v is restated in terms of ϕ , ensuring that both states always view the object as being worth fighting over.

Consider a transfer-protocol crisis bargaining game, as this provides a simple environment to isolate the amplification effect. Let $V_i(\theta_i; \lambda) = \int v(\theta_i, z) f_j(z | \theta_i; \lambda) dz$ denote state i of type θ_i 's interim expected stakes under dependence parameter λ . Then, we can derive a correlated-information analogue to Proposition 2's impossibility: if

$$c_0 + c_1 < p_j(V_j(1; \lambda) - V_i(-1; \lambda)) \quad (5)$$

for some state i and an opponent j , there does not exist a transfer-protocol crisis bargaining game with uncertain stakes that admits an always-peaceful equilibrium, under Assumptions 2 and 3. The formal statement is presented as Lemma S.9 (SI 1.11).

Correlation gives rise to two countervailing effects. The screening channel operates through the conditional density $f_j(\theta_j | \theta_i; \lambda)$, which places more weight on θ_j sharing the sign of θ_i . Under a

transfer protocol, incentive compatibility forces each state's interim transfer to be constant in its own type (Lemma S.7, SI 1.9), so what a state receives cannot be made to depend on what it reports. This effectively turns off the screening effect in transfer-protocol games.

The amplification effect can be seen in the extreme types. The most optimistic receiving type $\theta_j = 1$ comes to expect a higher opponent type θ_i , while the most reluctant paying type $\theta_i = -1$ grows equally sure its opponent's type θ_j is low. Because these extreme types bear the binding participation constraints (Lemma S.8, SI 1.10), what matters is the gap between the expected stakes of these extreme types, $V_j(1; \lambda) - V_i(-1; \lambda)$. This gap strictly widens in λ (Lemma S.10, SI 1.12), increasing the right-hand side of inequality (5). The total costs of war required to satisfy both extreme types therefore strictly rises with correlation.

Proposition 4. *Take any transfer-protocol crisis bargaining game with uncertainty about stakes, and let Assumptions 2 and 3 hold. If inequality (5) holds at $\lambda \in [0, 1]$ but not at $\tilde{\lambda} \in [0, 1]$, then it must be that $\lambda > \tilde{\lambda}$, and that inequality (5) holds at every $\lambda' > \lambda$ and does not hold at any $\lambda'' < \tilde{\lambda}$.*

The result qualifies the conventional logic that less uncertainty about an opponent's type promotes peace. Because a transfer protocol forces the interim transfer to be constant, the additional information correlation provides cannot be used to screen, and amplification causes the extreme types whose constraints bind to grow more confident that the stakes are extreme. As λ increases, so does the threshold in (5), and the disputes that transfers cannot consistently settle accumulate.

To illustrate, suppose the stakes are linear $\phi(\theta_0 + \theta_1) = \theta_0 + \theta_1 + k$ for some $k > 2 + \max_i c_i/p_i$ with evenly balanced power $p = 1/2$. The extreme high-type opponent j has an interim expected stakes of $V_j(1; \lambda) = 1 + k + \lambda/3$, and the extreme low-type state i has $V_i(-1; \lambda) = -1 + k - \lambda/3$.¹² Here, the amplification effect is clear: the gap between them, $2 + 2\lambda/3$, grows in λ , and always-peaceful equilibria are impossible if $c_0 + c_1 < 1 + \lambda/3$. As λ increases, always-peace requires even greater total costs of war.

6 Empirical illustrations

The results identify conditions under which peace cannot be guaranteed. Historical cases cannot verify claims of this kind, but they can show that the conditions arise in practice and that war and

¹² To see this, note that $E[\theta_i | \theta_j = 1] = (1/2) \int_{-1}^1 z(1 + \lambda z) dz = (\lambda/2) \int_{-1}^1 z^2 dz = \lambda/3$, and similarly for $E[\theta_j | \theta_i = -1]$.

peace have followed the general pattern described by the results. Moreover, while the impossibility results show that peace cannot be guaranteed under the conditions, they do not imply that war must occur (with probability one). It is entirely plausible, and does not contradict the paper's thesis, for cases to exist in which the conditions hold alongside peaceful agreement.

With this scope in mind, I present two brief empirical illustrations, one for each of the main theorems. First, the Mexican-American War presents a boundary indivisibility with uncertainty about stakes, where agreement could not be reached (Theorem 1). Second, the dispute between Ecuador and Peru over the Amazonian boundary presents an interior indivisibility, where agreement was reached but could not be maintained (Theorem 2).

Neither is a standard case in the literature on indivisible territory, likely because the contested objects themselves were divisible in the literal sense. The indivisibilities in these cases, as in the broader paper, lie in the feasible divisions of the object's value, of which the inability to divide an object physically is one source among others. Cases like these are brought into view by defining indivisibilities in this broader way, which includes, for example, the political feasibility of a settlement from the perspective of a decision-making leader.

Additionally, neither case is well explained by the standard causes of war. No commitment problem has been attributed to either. For Mexico, private information about power or the costs of war is a poor fit, as its leaders came to prefer even a war they expected to lose over a voluntary cession (Henderson, 2007). More favorable beliefs about the probability of victory or the costs of war would only raise the expected payoff from fighting relative to a given cession, pointing instead to a constraint on feasible peaceful concessions. For Ecuador and Peru, the balance of power was demonstrated through fighting, but the dispute remained unresolved until the settlement of 1998 severed access from sovereignty and changed what could be divided (Simmons, 1999, pp. 12, 20).

6.1 The Mexican-American War: Agreement could not be reached

When Polk's envoy John Slidell arrived in Mexico City in 1845, the two central features of Theorem 1 were in place. California's value was unknown to both sides, and an open cession of territory to the US was politically fatal for Mexican governments (Henderson, 2007). Mexico had held the territory for decades and learned little of what it contained, as it remained largely undeveloped, and American reports of its mineral wealth circulated but remained unconfirmed (Hammond, 1953).

Mexican President Herrera fell in December for his willingness even to receive Slidell, and his successor Paredes, having pledged to defend the territory, refused to receive him at all (Pletcher, 1973, pp. 364–377). With such divisions politically infeasible, the proposals that followed were built primarily on transfers. Slidell carried offers reaching \$25 million (Pletcher, 1973, pp. 288–291). Mexico, in turn, owed millions in defaulted claims to US citizens (Pletcher, 1973, p. 173), and settling them was its available means of compensating the US to keep the territory. The mission failed, and the US declared war in May 1846.

War as a result of uncertainty about the stakes is an especially compelling explanation for this case. On informational accounts, a war that revealed private information about power or the costs of war would have left one side accepting worse terms than it had initially refused. Instead, the Treaty of Guadalupe Hidalgo transferred the same territories on terms that were comparable to those the US offered before the war (Pletcher, 1973, pp. 500–501). Fighting revealed little if anything about the value of California. Whereas battle could inform beliefs on the balance of power or the costs of war, it is unlikely to reveal what the territory would prove to be worth. Within a year of signing the treaty, Polk reported to Congress that California’s mines were “more extensive and valuable than was anticipated” (Richardson, 1897, vol. 4).

In contrast to these events, the Gadsden Purchase six years later shows transfers succeeding once the stakes were known. In 1854, Mexico accepted \$10 million for a strip surveyed as the route for a southern railroad, whose value both sides could reliably assess (Garber, 1923). Unlike before, transfers could substitute for the territory at a price both sides accepted.

6.2 The Cenepa War: Agreement could not be maintained

For most of the twentieth century, Ecuador and Peru disputed their boundary in the upper Amazon. When they settled it in the 1942 Rio Protocol, the two main conditions of Theorem 2 were present. First, there was uncertainty about the value of the territory at stake, which was long suspected to hold oil. Indeed, the suspicion was strong enough that companies, including Royal Dutch Shell, had paid to explore it for decades, yielding only “undetermined results” (Alarcón, 2021, p. 101).

Second, while the *territory* was divisible, settlements at some interior divisions of *what was at stake* were not feasible. Specifically, the dispute was never just about the land. Ecuador sought sovereign access to the Amazon, whose river system would connect its interior to Atlantic trade

without depending on Peru's consent. Either a settlement provided Ecuador with sovereign access to the river or it did not. No division could deliver only part of that access, and whichever side controlled the passage would collect what the position proved to be worth (Mares and Palmer, 2012).

The 1942 Protocol nevertheless allowed the two sides to reach an agreement before the precise implications of the boundary were known. Rather than draw the entire line on a map, the Protocol described it using geographic features, including the divide between the Zamora and Santiago rivers, with the precise boundary still to be determined on the ground (Simmons, 1999, pp. 10, 26–27). The parties therefore agreed to a rule for locating the boundary without knowing exactly what territorial division that rule would ultimately produce. Agreement could be reached on those terms, but whether it would endure depended on whether the resulting boundary remained acceptable to both sides.

The boundary that was originally agreed to, however, turned out not to be mutually acceptable. In 1948, Ecuador halted demarcation, asserting that aerial photography had revealed a river system more extensive than previously understood. The line that was agreed to left Ecuador without sovereign access to the Amazon, and in 1960 Ecuador declared the Protocol null and void (Simmons, 1999, p. 10). Peru rejected Ecuador's claim, insisting that the Protocol remained in force and that the boundary dispute had already been settled (Simmons, 1999, p. 11). The dispute eventually returned to war in the Cenepa valley in 1995, but fighting did not resolve the disagreement that had caused the Protocol to break down. Although Ecuadorian forces achieved some success on the ground, the postwar settlement continued to recognize the Protocol boundary (Simmons, 1999, pp. 12, 20–21). The sequence points to uncertainty about stakes interacting with the interior indivisibility, as fighting can reveal information about the balance of power or the costs of war, but it does not change what could be divided. Ecuador either obtained sovereign access to the Amazon or it did not, and intermediate territorial divisions could not provide intermediate amounts of that value.

The 1998 Brasília accords resolved the dispute without changing the Protocol boundary. Peru retained sovereignty over the disputed territory, while Ecuador received navigation rights on the Amazon and property without sovereignty at Tiwintza (Simmons, 1999, p. 20). By separating access and property rights from sovereignty, the settlement allowed Ecuador to obtain some of the value it sought from the disputed territory without obtaining sovereign control over it. This became credible in 1998 when, after bilateral negotiations reached another impasse, Ecuador and Peru agreed to

binding arbitration by Argentina, Brazil, Chile, and the United States (Simmons, 1999, pp. 20, 22). In terms of the theoretical framework, the change in the bargaining institution expanded the feasible shares of the object's value that could be credibly implemented without changing the territorial boundary.

7 Conclusion

This paper examines crisis bargaining games when there is an unknown common value for the object, which I call the stakes of the dispute. I develop a new formulation of the crisis bargaining game form to study every possible game in this class, and show that the interaction of uncertainty about the stakes of the dispute with restrictions on peaceful divisions of the object's value can make it impossible to always avoid war. This holds true even as states are unrestricted in the transfers they can make to reach a peaceful agreement and hence there is no indivisibility in the surplus from peace.

Specifically, I show that states are unable to always reach peaceful agreements if extreme divisions are ruled out (Theorem 1). I illustrate the intuition behind this result using benchmark transfer-protocol subclass in which peaceful settlements rely exclusively on transfers (Proposition 2). Additionally, when interior divisions are precluded instead, peaceful agreements that randomize over settlements always exist (Proposition 3), but every such agreement relies on commitments that states cannot enforce on their own (Theorem 2). This runs against conventional wisdom that suggests transfers in the form of side payments and concessions on other issues can rescue peaceful settlements in the face of indivisibilities. Finally, I consider an extension in which states have correlated private information. I identify countervailing screening and amplification effects, derive an impossibility condition in a special case, and show that correlation can make always-peace even more difficult to achieve (Proposition 4).

The findings have implications for what types of disputes are hardest to resolve peacefully. If negotiating a settlement requires an exchange of transfers, such as a concession on another issue, then the resolution procedures are especially susceptible to failure. This is because states cannot denominate transfers in the unknown value of the object, and so any state compensated with transfers either misses out on upside when the stakes are high, or invites lower types to mimic their report. Extending that logic, even small indivisibilities can pose a significant barrier to peace, especially when

there is a highly asymmetric power balance. Finally, the results have implications for institutional design: institutions that seek to improve information can counterintuitively undermine the prospects for peace, as states with extreme beliefs about the stakes come to expect their opponents to share them, leaving both sides less willing to compromise.

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A Preliminaries: Crisis bargaining game form, direct mechanisms, and always-peaceful equilibria

In this section, I define the class of crisis bargaining games and present the results that connect their equilibria to direct mechanisms. The definition leaves actions and timing of the game unrestricted and imposes only the substantive features of crisis bargaining, in which states may peacefully divide an object and exchange transfers to settle the dispute, and no state can be compelled to accept less than it would expect from war. Unlike previous work, I leave the object's value unnormalized, allow for mixed strategies and stochastic agreements, and distinguish shares from transfers as settlement instruments. The latter proves especially consequential when the value of the contested object is unknown. In the main text, I restrict attention to games with uncertainty about stakes.

A crisis bargaining game Γ is an interaction between two states $i = 0, 1$. Throughout the paper, I use j to denote a state i 's opponent. Each state chooses actions $a = (a_0, a_1) \in A \equiv A_0 \times A_1$, which lead to a final outcome of war or peaceful settlement. In war, each state incurs costs $c_i > 0$ to fight for an object they value $v_i > 0$, which they win with probability $p_i = 1 - p_j$. Both probabilities are pinned down by $p = p_0$. Therefore, we can denote each state's war payoff by $w_i = v_i p_i - c_i$. On the other hand, peaceful settlements can include two components. First, states may agree to a share or division of the object, $x_i(a) \in [0, 1]$. For a state i to receive a share implies their opponent receives the remainder, hence $x_i(a) = 1 - x_j(a)$ for all $a \in A$. Second, states may extend and receive transfers in the form of side payments or compromises on other issues, netting to a gain of $t_i(a) \in \mathbb{R}$. For a state i to extend a transfer implies their opponent j receives it, hence $t_i(a) + t_j(a) = 0$ for all $a \in A$. From this, state i 's payoff from cooperating under action profile a can be expressed by $v_i x_i(a) + t_i(a)$.

A crisis bargaining game Γ concludes with an outcome that can be represented by $(\pi, x, t) \in \Omega \equiv \{0, 1\} \times [0, 1] \times \mathbb{R}$, where π indicates whether war occurs and, if war does not occur, x yields the share allocated to state 0 while t yields the transfer allocated to state 0. Then, we can say that any crisis bargaining game has a corresponding outcome function $\gamma : A \rightarrow \Delta(\Omega)$ that takes actions and yields a distribution over outcomes. For any action profile, the induced moments are given by $\pi^\gamma(a) \equiv E_{\gamma(a)}\pi$ as the probability of war and, in a peaceful settlement, $x^\gamma(a) \equiv E_{\gamma(a)}[x \mid \pi = 0]$ is the expected share of object allocated to state 0 while $t^\gamma(a) \equiv E_{\gamma(a)}[t \mid \pi = 0]$ is the expected transfer

received by 0. Then, the expected utility for a state i under an action profile a is

$$u_i(a) = \pi^\gamma(a)w_i + (1 - \pi^\gamma(a))(v_i x_i^\gamma(a) + t_i^\gamma(a)) \quad (\text{A.1})$$

where $x^\gamma(a) = x_0^\gamma(a) = 1 - x_1^\gamma(a)$ and $t^\gamma(a) = t_0^\gamma(a) = -t_1^\gamma(a)$. When $\pi^\gamma(a) = 1$, the peace-conditioned moments may be set arbitrarily, as they enter payoffs only through the factor $1 - \pi^\gamma(a)$. The same convention applies to the direct-mechanism moments.

It is now time to incorporate private information. Let each state have a private type $\theta_i \in \Theta_i$ with $\Theta_i \subset \mathbb{R}$ a non-degenerate compact interval. A type pair $\theta = (\theta_0, \theta_1)$ is drawn from common joint prior F on $\Theta \equiv \Theta_0 \times \Theta_1$. Private information may concern any collection of the primitives (v_0, v_1, p, c_0, c_1) , though subsequent sections will focus on the case where p , c_0 , and c_1 are common knowledge. When a primitive depends on types, we can write i 's expected utility given θ as $u_i(a, \theta)$ to make this dependence explicit, with war payoffs likewise expressed as $w_i(\theta)$. Letting each state's strategy be given by $\sigma_i : \Theta_i \rightarrow \Delta(A_i)$, a strategy profile $\sigma = (\sigma_0, \sigma_1)$ induces a distribution $\sigma(\theta) \in \Delta(A)$ over action profiles at each type pair. The interim expected utility for a state of type θ_i given a strategy profile σ can then be written as

$$U_i^\sigma(\theta_i) = \int_{\Theta_j} E_{\sigma(\theta_i, z)} u_i(a, (\theta_i, z)) dF(z | \theta_i). \quad (\text{A.2})$$

The only assumption we require on the action space is that it satisfies what is known as the voluntary agreements (VA) criterion. Due to the anarchic nature of crisis bargaining, it is standard to focus on settings where states can always unilaterally guarantee themselves an expected payoff of at least their war payoff. I thus make the following assumption, which guarantees each state its war payoff in expectation only. Accordingly, it is weaker than the typical voluntary agreements criterion, placing no restriction on realized settlement terms.

Assumption A.1 (Voluntary Agreements). *For any crisis bargaining game Γ , there exists an action $\tilde{a}_i \in A_i$ for each state i such that, for every measurable $a_j : \Theta_j \rightarrow A_j$ and all $\theta_i \in \Theta_i$,*

$$E_{\theta_j} [u_i((\tilde{a}_i, a_j(\theta_j)), \theta) | \theta_i] \geq E_{\theta_j} [w_i(\theta) | \theta_i]. \quad (\text{VA})$$

Finally, a direct mechanism $\delta : \Theta \rightarrow \Delta(\Omega)$ is a function mapping type profiles to distributions

of outcomes. Fix a Bayesian Nash equilibrium σ^* of a crisis bargaining game Γ , then δ is called an *equivalent direct mechanism* if it yields the same distribution of outcomes induced by σ^* and outcome function γ . A simple way to think about this is that there is a new game in which states only report a type to a mechanism designer, who then selects an outcome those types would have obtained in the original game. Then, for any reported type profile θ , the corresponding moments are given by $\pi^\delta(\theta) = E_{\delta(\theta)}\pi$ for the probability of war and $(x^\delta, t^\delta)(\theta) = E_{\delta(\theta)}[(x, t) \mid \pi = 0]$ for the expected share and transfer in peace. The interim expected utility of a state with type θ_i reporting as type $\tilde{\theta}_i$ under strategies σ^* can now be expressed in terms of the equivalent direct mechanism by

$$U_i^\delta(\tilde{\theta}_i; \theta_i) = \int_{\Theta_j} u_i^\delta(\tilde{\theta}_i; (\theta_i, z)) dF(z \mid \theta_i), \text{ where}$$

$$u_i^\delta(\tilde{\theta}_i; (\theta_i, \theta_j)) = \pi^\delta(\tilde{\theta}_i, \theta_j) w_i(\theta_i, \theta_j) + (1 - \pi^\delta(\tilde{\theta}_i, \theta_j)) (v_i(\theta_i, \theta_j) x_i^\delta(\tilde{\theta}_i, \theta_j) + t_i^\delta(\tilde{\theta}_i, \theta_j)),$$

and likewise $x^\delta(\theta) = x_0^\delta(\theta) = 1 - x_1^\delta(\theta)$ and $t^\delta(\theta) = t_0^\delta(\theta) = -t_1^\delta(\theta)$ always hold.

By definition of an equivalent direct mechanism, truth-telling in the direct mechanism yields equivalent expected utility as equilibrium play in the original game, i.e., for all θ_i , $U_i^\delta(\theta_i; \theta_i) = U_i^{\sigma^*}(\theta_i)$. Now, we have the machinery necessary to close this section with brief remarks on the mechanism design result that makes the remainder of the analysis possible.

Definition A.1 (Incentive Compatibility). *A direct mechanism δ is called incentive compatible if and only if*

$$U_i^\delta(\theta_i; \theta_i) \geq U_i^\delta(\tilde{\theta}_i; \theta_i) \tag{IC}$$

for each state i , for all $\theta_i, \tilde{\theta}_i \in \Theta_i$.

Remark A.1 (Myerson (1979)). *If σ^* is a Bayesian Nash equilibrium of a crisis bargaining game Γ , there exists an incentive-compatible direct mechanism δ that induces the same distribution over outcomes as outcome function γ and equilibrium strategies σ^* .*

This result, known as the revelation principle, is exceptional in that it facilitates the analysis of broad classes of games through an investigation of direct mechanisms. In particular, because there must exist an incentive-compatible direct mechanism that produces the same outcomes for any

equilibrium of a game in this class, we can make positive statements about properties of equilibria based on the outcomes that can be produced by incentive-compatible direct mechanisms.

The converse also holds but requires the condition $U_i^\delta(\theta_i; \theta_i) \geq E_{\theta_j}[w_i(\theta) | \theta_i]$ for each state i and every type θ_i , which captures the equilibrium implications of the VA criterion on the direct mechanism δ (Lemma S.1, SI 1.1). Accordingly, to establish the existence of a crisis bargaining game that induces a particular distribution over outcomes, it suffices to exhibit such an incentive-compatible direct mechanism that induces that same distribution, and vice versa. In this paper, as in previous game-free analyses of crisis bargaining, I focus on the existence of games that admit always-peaceful equilibria.

Definition A.2 (Always-Peaceful Equilibrium). *An equilibrium σ^* of a crisis bargaining game Γ is always peaceful if and only if $\pi = 0$ for all $(\pi, x, t) \in \text{supp}(\gamma(a))$, $a \in \text{supp}(\sigma^*(\theta))$, and $\theta \in \Theta$.*

Throughout the analysis, a positive probability of war means that war occurs with positive probability at some type profile under equilibrium play. Thus, always-peace requires this probability to be zero at every type profile.

Remark A.2. *There exists a crisis bargaining game that admits an always-peaceful equilibrium if and only if there exists an incentive-compatible direct mechanism δ such that $\pi = 0$ for all $(\pi, x, t) \in \text{supp}(\delta(\theta))$ and all $\theta \in \Theta$, and $U_i^\delta(\theta_i; \theta_i) \geq E_{\theta_j}[w_i(\theta) | \theta_i]$ for each state i and all $\theta_i \in \Theta_i$.*

The forward direction holds because deviating to $\tilde{a}_i$ against equilibrium play yields at least the interim war payoff by Assumption A.1. The equivalent direct mechanism of Remark A.1 induces the same payoffs, so it satisfies the same bound and never produces war. The converse follows from Lemma S.1. Therefore, to recover such an incentive-compatible direct mechanism that produces an outcome that is always peaceful is equivalent to recovering a crisis bargaining game that admits a Bayesian Nash equilibrium that is always peaceful.

B Indivisibilities with known stakes

This section provides a more formal statement of the benchmark result with known stakes, corresponding to Proposition 1. The statement is on the full class of crisis bargaining games, in which power p and the costs of war (c_0, c_1) may be unknown.

Proposition B.1. *Suppose the object's value is common to both states and commonly known, $v_0 = v_1 = v$, while power p and the costs of war (c_0, c_1) may be unknown. Fix any nonempty set of feasible divisions $X \subseteq [0, 1]$ and any incentive-compatible direct mechanism δ . Then there exists an incentive-compatible direct mechanism $\tilde{\delta}$ such that $x \in X$ for all $(\pi, x, t) \in \text{supp}(\tilde{\delta}(\theta))$ and all $\theta \in \Theta$, with the same probability of war and the same ex-post payoffs as δ at every type profile. Moreover, $\tilde{\delta}$ is robustly implementable whenever δ is, and if $U_i^\delta(\theta_i; \theta_i) \geq E_{\theta_j}[w_i(\theta) | \theta_i]$ for each state i and all $\theta_i \in \Theta_i$, there exists a crisis bargaining game whose feasible divisions lie in X with an equilibrium of which $\tilde{\delta}$ is an equivalent direct mechanism.*

The proof of this statement is presented in the Supplemental Information (SI 3.1).

Supplemental Information for “Crisis Bargaining Over What?”

For online publication only

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SI 1 Auxiliary results

The paper's main results depend on the following auxiliary results. Lemma S.1 applies broadly to every crisis bargaining game with uncertainty about any collection of primitives (v_0, v_1, p, c_0, c_1) . In all other statements, I assume a crisis bargaining game with uncertain stakes such that power p and costs of war (c_0, c_1) are common knowledge, but the value of the object is common $v_0 = v_1 = v$ and unknown to both sides.

All formal statements are numbered by the order in which they are presented. To make the logical structure between the main and auxiliary results explicit, Proposition B.1 and 1 can be proved after Lemma S.1, Propositions S.1 and 3 after Lemma S.2, Theorem 1 and Proposition 2 after Lemma S.4, Theorem 2 after Lemma S.6, and Proposition 4 after Lemma S.10.

SI 1.1 Lemma S.1

By the revelation principle, the existence of an incentive-compatible direct mechanism that induces a distribution over outcomes is implied by the existence of a Bayesian Nash equilibrium that induces that distribution. To establish the converse, we additionally require the following result.

Lemma S.1. *Let δ be an incentive-compatible direct mechanism such that $U_i^\delta(\theta_i; \theta_i) \geq E_{\theta_j}[w_i(\theta) | \theta_i]$ for each state i and all $\theta_i \in \Theta_i$. Then, there exists a crisis bargaining game Γ and a Bayesian Nash equilibrium σ^* of Γ of which δ is an equivalent direct mechanism.*

Proof. Take any incentive-compatible direct mechanism δ such that $U_i^\delta(\theta_i; \theta_i) \geq E_{\theta_j}[w_i(\theta) | \theta_i]$ for each state i and all $\theta_i \in \Theta_i$. Construct the following game.

Let each state i have actions $A_i = \Theta_i \cup \{W_i\}$ where W_i is state i 's action to fight a war, and define the outcome function by $\gamma(a) = \delta(a)$ if $a \in \Theta$ and $\gamma(a)(\{(1, x, t)\}) = 1$ for any $(x, t) \in [0, 1] \times \mathbb{R}$ otherwise. Let the war action by either state yield war payoffs against any action of their opponent, therefore satisfying the VA criterion of Assumption A.1.

Now consider $\sigma^*(\theta) = \theta$. Fix a state i , type θ_i , and a deviation $\tilde{a}_i \in A_i$. If $\tilde{a}_i = \tilde{\theta}_i \in \Theta_i$, then the deviation yields $U_i^\delta(\tilde{\theta}_i; \theta_i) \leq U_i^\delta(\theta_i; \theta_i)$ by incentive compatibility of δ . If $\tilde{a}_i = W_i$, then the deviation yields $E_{\theta_j}[w_i(\theta) | \theta_i] \leq U_i^\delta(\theta_i; \theta_i)$ by the antecedent. Thus, σ^* is a Bayesian Nash equilibrium and $\gamma \circ \sigma^*$ induces the same distribution over outcomes as δ . $\square$

SI 1.2 Lemma S.2

For any crisis bargaining game Γ , let $\mu_i(\theta_i)$ denote a Bayesian posterior belief of state i with type θ_i at some point along the path of play. Suppose we were to impose a stronger notion of the VA criterion such that states can guarantee at least their war payoff upon learning this information, so that a direct mechanism δ must satisfy

$$\mathbb{E}_{\theta_j \sim \mu_i(\theta_i)} [v(\theta_i, \theta_j) x_i^\delta(\theta_i, \theta_j) + t_i^\delta(\theta_i, \theta_j) - w_i(\theta_i, \theta_j)] \geq 0 \quad (\text{S.1})$$

for each state i , all $\theta_i \in \Theta_i$, and every posterior belief realized along the path of play.

Inequality (S.1) reflects a participation constraint for peaceful settlement that depends on game-specific posteriors. The following lemma reduces this posterior condition to an interim participation constraint.

Lemma S.2 (Interim Participation Constraint). *If a direct mechanism δ satisfies $\pi = 0$ for all $(\pi, x, t) \in \text{supp}(\delta(\theta))$, $\theta \in \Theta$, and inequality (S.1) holds for each state i , all $\theta_i \in \Theta_i$, and every realized posterior belief, then $U_i^\delta(\theta_i; \theta_i) \geq V_i(\theta_i)p_i - c_i$ for each state i and all $\theta_i \in \Theta_i$.*

Proof. Fix a state i of type θ_i . Taking expectations over the realized posterior $\mu_i(\theta_i)$ given θ_i ,

$$\mathbb{E}_{\mu_i(\theta_i)} [\mathbb{E}_{\theta_j \sim \mu_i(\theta_i)} [v(\theta) x_i^\delta(\theta) + t_i^\delta(\theta) - w_i(\theta)]] = \mathbb{E}_{\theta_j} [v(\theta) x_i^\delta(\theta) + t_i^\delta(\theta) - w_i(\theta) \mid \theta_i]$$

by the law of iterated expectations. Inequality (S.1) therefore implies

$$\mathbb{E}_{\theta_j} [v(\theta) x_i^\delta(\theta) + t_i^\delta(\theta) \mid \theta_i] \geq \mathbb{E}_{\theta_j} [w_i(\theta) \mid \theta_i]. \quad (\text{S.2})$$

By definition, the right-hand side of inequality (S.2) is equal to $V_i(\theta_i)p_i - c_i$ and, because $\pi = 0$ for all $(\pi, x, t) \in \text{supp}(\delta(\theta))$ and all $\theta \in \Theta$, the left-hand side of (S.2) is $U_i^\delta(\theta_i; \theta_i)$. $\square$

SI 1.3 Corollary S.1

Corollary S.1. *Suppose F admits a conditional density $f_j(\cdot | \theta_i)$ that is continuously differentiable in (θ_i, θ_j) . In any always-peaceful equilibrium σ^* of a share-protocol crisis bargaining game,*

$$\frac{dU_i^{\sigma^*}(\theta_i)}{d\theta_i} = \int_{\Theta_j} \frac{\partial v}{\partial \theta_i}(\theta_i, z) x_i^\delta(\theta_i, z) f_j(z | \theta_i) dz + \int_{\Theta_j} v(\theta_i, z) x_i^\delta(\theta_i, z) \frac{\partial f_j(z | \theta_i)}{\partial \theta_i} dz$$

for each state i and almost all $\theta_i \in \Theta_i$.

Proof. In any always-peaceful equilibrium of a share-protocol game, $\pi^\delta(\theta) = 0$ and $t^\delta(\theta) = 0$ for all $\theta \in \Theta$, so $U_i^\delta(\tilde{\theta}_i; \theta_i) = \int_{\Theta_j} v(\theta_i, z) x_i^\delta(\tilde{\theta}_i, z) dF_j(z | \theta_i)$. Because f_j is C^1 in θ_i , the partial derivative is

$$\frac{\partial U_i^\delta(\tilde{\theta}_i; \theta_i)}{\partial \theta_i} = \int_{\Theta_j} \frac{\partial v}{\partial \theta_i}(\theta_i, z) x_i^\delta(\tilde{\theta}_i, z) f_j(z | \theta_i) dz + \int_{\Theta_j} v(\theta_i, z) x_i^\delta(\tilde{\theta}_i, z) \frac{\partial f_j(z | \theta_i)}{\partial \theta_i} dz. \quad (\text{S.3})$$

By incentive compatibility, $U_i^{\sigma^*}(\theta_i) = \sup_{\tilde{\theta}_i} U_i^\delta(\tilde{\theta}_i; \theta_i)$ and hence, by the envelope theorem (Milgrom and Segal, 2002, Theorem 2),

$$\frac{dU_i^{\sigma^*}(\theta_i)}{d\theta_i} = \left. \frac{\partial U_i^\delta(\tilde{\theta}_i; \theta_i)}{\partial \theta_i} \right|_{\tilde{\theta}_i = \theta_i}. \quad (\text{S.4})$$

Evaluating equation (S.3) at $\tilde{\theta}_i = \theta_i$ and substituting into (S.4) yields the desired expression. $\square$

SI 1.4 Corollary S.2

Corollary S.2. *Suppose Assumption 1 holds and fix state i so that $x_i = 1$ for all $(0, x, t) \in \text{supp}(\delta(\theta))$, $\theta \in \Theta$. Then, in any equilibrium σ^* of a transfer-protocol crisis bargaining game, for almost all types,*

$$\begin{aligned} \frac{dU_i^{\sigma^*}(\theta_i)}{d\theta_i} &= \int_{\Theta_j} \frac{\partial v}{\partial \theta_i}(\theta_i, z) (1 - p_j \pi^\delta(\theta_i, z)) dF_j(z) \\ \frac{dU_j^{\sigma^*}(\theta_j)}{d\theta_j} &= p_j \int_{\Theta_i} \frac{\partial v}{\partial \theta_j}(\theta_j, z) \pi^\delta(\theta_j, z) dF_i(z). \end{aligned}$$

Proof. Fix a state i such that $x_i^\delta = 1$ in any peaceful settlement and note that, by Assumption 1, $dF(z | \theta_i) = dF_j(z)$ and $dF(z | \theta_j) = dF_i(z)$.

First, taking the partial derivative of $U_i^\delta(\tilde{\theta}_i; \theta_i)$ with respect to θ_i , we have

$$\frac{\partial U_i^\delta(\tilde{\theta}_i; \theta_i)}{\partial \theta_i} = \int_{\Theta_j} \frac{\partial v}{\partial \theta_i}(\theta_i, z)(1 - p_j \pi^\delta(\tilde{\theta}_i, z)) dF_j(z). \quad (\text{S.5})$$

By incentive compatibility $U_i^{\sigma^*}(\theta_i) = \sup_{\tilde{\theta}_i} U_i^\delta(\tilde{\theta}_i; \theta_i)$, and the envelope theorem (Milgrom and Segal, 2002, Theorem 2) gives $dU_i^{\sigma^*}/d\theta_i = \partial U_i^\delta/\partial \theta_i|_{\tilde{\theta}_i=\theta_i}$. Then, evaluating equation (S.5) at $\tilde{\theta}_i = \theta_i$ recovers the first expression.

Second, taking the partial derivative of $U_j^\delta(\tilde{\theta}_j; \theta_j)$ with respect to θ_j ,

$$\frac{\partial U_j^\delta(\tilde{\theta}_j; \theta_j)}{\partial \theta_j} = p_j \int_{\Theta_i} \frac{\partial v}{\partial \theta_j}(\theta_j, z) \pi^\delta(\tilde{\theta}_j, z) dF_i(z). \quad (\text{S.6})$$

By the same envelope theorem argument, evaluating (S.6) at $\tilde{\theta}_j = \theta_j$ yields the second expression. $\square$

SI 1.5 Lemma S.3

Lemma S.3. *If Assumption 1 holds, then for any equilibrium σ^* of any crisis bargaining game Γ with incentive-compatible equivalent direct mechanism δ , the interim expected utility for state with type θ_i must satisfy, for almost all $\theta_i \in \Theta_i$,*

$$\frac{dU_i^{\sigma^*}(\theta_i)}{d\theta_i} = \int_{\Theta_j} \frac{\partial v}{\partial \theta_i}(\theta_i, z) [\pi^\delta(\theta_i, z) p_i + (1 - \pi^\delta(\theta_i, z)) x_i^\delta(\theta_i, z)] dF_j(z).$$

If σ^ is an always-peaceful equilibrium, $\pi^\delta(\theta) = 0$ for all $\theta \in \Theta$ and therefore*

$$\frac{dU_i^{\sigma^*}(\theta_i)}{d\theta_i} = \int_{\Theta_j} \frac{\partial v}{\partial \theta_i}(\theta_i, z) x_i^\delta(\theta_i, z) dF_j(z).$$

Proof. By incentive compatibility, $U_i^{\sigma^*}(\theta_i) = U_i^\delta(\theta_i; \theta_i) = \sup_{\tilde{\theta}_i \in \Theta_i} U_i^\delta(\tilde{\theta}_i; \theta_i)$. Because v is C^1 on compact Θ , $|\partial v/\partial \theta_i|$ is uniformly bounded in $\tilde{\theta}_i$, so $U_i^\delta(\tilde{\theta}_i; \theta_i)$ is absolutely continuous on Θ_i . By the envelope theorem (Milgrom and Segal, 2002, Theorem 2), we then have

$$\frac{dU_i^{\sigma^*}(\theta_i)}{d\theta_i} = \left. \frac{\partial U_i^\delta(\tilde{\theta}_i; \theta_i)}{\partial \theta_i} \right|_{\tilde{\theta}_i=\theta_i}.$$

Moreover,

$$\begin{aligned}\frac{\partial U_i^\delta(\tilde{\theta}_i; \theta_i)}{\partial \theta_i} &= \int_{\Theta_j} \frac{\partial u_i^\delta}{\partial \theta_i}(\tilde{\theta}_i; (\theta_i, z)) dF_j(z) \\ &= \int_{\Theta_j} \frac{\partial v}{\partial \theta_i}(\theta_i, z) [\pi^\delta(\tilde{\theta}_i, z) p_i + (1 - \pi^\delta(\tilde{\theta}_i, z)) x_i^\delta(\tilde{\theta}_i, z)] dF_j(z)\end{aligned}$$

by Assumption 1 and Leibniz's rule. Evaluating the expression at $\tilde{\theta}_i = \theta_i$ yields the stated result. The expression for an always-peaceful equilibrium follows directly by plugging in zero probability of war. $\square$

SI 1.6 Lemma S.4

Lemma S.4. *Suppose that Assumption 1 holds and fix an always-peaceful equilibrium σ^* of a crisis bargaining game Γ that has an incentive-compatible equivalent direct mechanism δ . If $x_i^\delta(\theta) \geq \bar{x}$ for a state i and all $\theta \in \Theta$, then the bounds*

$$V_i(\theta_i) - \underline{V}_i \geq U_i^{\sigma^*}(\theta_i) - U_i^{\sigma^*}(\underline{\theta}_i) \geq \bar{x}(V_i(\theta_i) - \underline{V}_i) \quad (\text{S.7})$$

$$\text{and} \quad (1 - \bar{x})(V_j(\theta_j) - \underline{V}_j) \geq U_j^{\sigma^*}(\theta_j) - U_j^{\sigma^*}(\underline{\theta}_j) \geq 0 \quad (\text{S.8})$$

must hold for all $\theta_i \in \Theta_i$ and $\theta_j \in \Theta_j$.

Proof. Recall that for each state,

$$\frac{dU_i(\theta_i)}{d\theta_i} = \int_{\Theta_j} \frac{\partial v}{\partial \theta_i}(\theta_i, z) x_i^\delta(\theta_i, z) dF_j(z) \quad (\text{S.9})$$

by Lemma S.3. Fix a state i and opponent j such that $x_i^\delta(\theta) = 1 - x_j^\delta(\theta) \geq \bar{x}$ for all $\theta \in \Theta$.

Then, first consider equation (S.9) for state i . Because $1 \geq x_i^\delta(\theta) \geq \bar{x}$ for all θ and $\partial v / \partial \theta_i \geq 0$, the integrand can be bounded as

$$\frac{\partial v}{\partial \theta_i}(\theta_i, \theta_j) \geq \frac{\partial v}{\partial \theta_i}(\theta_i, \theta_j) x_i^\delta(\theta_i, \theta_j) \geq \frac{\partial v}{\partial \theta_i}(\theta_i, \theta_j) \bar{x}$$

for all $\theta_j \in \Theta_j$. Integrating both sides, we recover

$$\int_{\Theta_j} \frac{\partial v}{\partial \theta_i}(\theta_i, z) dF_j(z) \geq \frac{dU_i^{\sigma^*}(\theta_i)}{d\theta_i} \geq \bar{x} \int_{\Theta_j} \frac{\partial v}{\partial \theta_i}(\theta_i, z) dF_j(z).$$

By Assumption 1, Leibniz's rule, and the continuous differentiability of v , we can simplify the expression to $V_i'(\theta_i) \geq dU_i^{\sigma^*}(\theta_i)/d\theta_i \geq \bar{x}V_i'(\theta_i)$. Fixing θ_i and integrating with respect to Lebesgue measure from $\underline{\theta}_i$ to θ_i then yields

$$\int_{\underline{\theta}_i}^{\theta_i} V_i'(z) dz \geq \int_{\underline{\theta}_i}^{\theta_i} \frac{dU_i^{\sigma^*}}{dz}(z) dz \geq \bar{x} \int_{\underline{\theta}_i}^{\theta_i} V_i'(z) dz.$$

Because $U_i^{\sigma^*}$ and V_i are absolutely continuous, we have equation (S.7) by the fundamental theorem of calculus.

We derive bounds for opponent j by analogous argument. Because $1 - \bar{x} \geq x_j^\delta(\theta) \geq 0$, we have

$$\frac{\partial v}{\partial \theta_j}(\theta_i, \theta_j)(1 - \bar{x}) \geq \frac{\partial v}{\partial \theta_j}(\theta_i, \theta_j)x_j^\delta(\theta_i, \theta_j) \geq 0$$

for all $\theta_i \in \Theta_i$, from which the identical steps as above imply equation (S.8). $\square$

SI 1.7 Lemma S.5

Lemma S.5. *Fix a direct mechanism δ such that $\pi^\delta(\theta) = 0$ for all $\theta \in \Theta$. If δ satisfies ex-post incentive compatibility and v is strictly increasing in both of its arguments, then $x^\delta(\theta_0, \theta_1)$ is non-decreasing in θ_0 and non-increasing in θ_1 .*

If δ is robustly implementable and $x^\delta(\theta) \notin (\underline{x}, \bar{x})$ for all $\theta \in \Theta$ for some $\underline{x}$ and $\bar{x}$ with $1 > \bar{x} > \underline{x} > 0$, there exists a non-decreasing threshold $\theta_i^(\theta_j)$ for each state i and any $\theta_j \in \Theta_j$ such that $x_i^\delta(\theta_i, \theta_j) \leq \underline{x}^{1-i}(1 - \bar{x})^i$ for all $\theta_i < \theta_i^*(\theta_j)$ and $x_i^\delta(\theta_i, \theta_j) \geq \bar{x}^{1-i}(1 - \underline{x})^i$ for all $\theta_i > \theta_i^*(\theta_j)$.*

Proof. Fix any $\theta_i, \tilde{\theta}_i \in \Theta_i$ and $\theta_j \in \Theta_j$ such that $\theta_i > \tilde{\theta}_i$. By ex-post incentive compatibility, we know that

$$(v(\theta_i, \theta_j) - v(\tilde{\theta}_i, \theta_j))(x_i^\delta(\theta_i, \theta_j) - x_i^\delta(\tilde{\theta}_i, \theta_j)) \geq 0.$$

By v strictly increasing, it must be that x_i^δ is non-decreasing in θ_i and, by symmetric argument,

non-increasing in θ_j .

Now suppose that there is an interior indivisibility and δ is robustly implementable. Then, for all θ , $\text{supp}(\delta(\theta)) = \{(0, x^\delta(\theta), t^\delta(\theta))\}$ and, by Definitions 3 and 4, $x^\delta(\theta) \notin (\underline{x}, \bar{x})$. Then, if for a fixed θ_j , $x_i^\delta(\theta) \leq \underline{x}^{1-i}(1 - \bar{x})^i$ for all $\theta_i \in \Theta_i$, the statement holds with $\theta_i^*(\theta_j) = \bar{\theta}_i$.¹³ Similarly, if $x_i^\delta(\theta) \geq \bar{x}^{1-i}(1 - \underline{x})^i$ for all $\theta_i \in \Theta_i$, the statement holds with $\theta_i^*(\theta_j) = \underline{\theta}_i$.

If, however, there exist $\theta_i, \tilde{\theta}_i \in \Theta_i$ such that $x_i^\delta(\theta_i, \theta_j) \geq \bar{x}^{1-i}(1 - \underline{x})^i > \underline{x}^{1-i}(1 - \bar{x})^i \geq x_i^\delta(\tilde{\theta}_i, \theta_j)$ for a given θ_j , then we must show a unique threshold in state i 's type. By fixing any θ_j and supposing that the above inequality holds for some types $\tilde{\theta}_i > \theta_i$, we arrive immediately at a contradiction to x_i^δ non-decreasing in θ_i .

Finally, i 's threshold $\theta_i^*(\theta_j)$ must be non-decreasing in θ_j . To see this, fix any $\theta_j > \tilde{\theta}_j$. We have established that x_i^δ is non-increasing in θ_j and thus, for any $\theta_i \in \Theta_i$, we must have $x_i^\delta(\theta_i, \theta_j) \leq x_i^\delta(\theta_i, \tilde{\theta}_j)$. Therefore, $x_i^\delta(\theta_i, \theta_j) \geq \bar{x}^{1-i}(1 - \underline{x})^i$ implies $x_i^\delta(\theta_i, \tilde{\theta}_j) \geq \bar{x}^{1-i}(1 - \underline{x})^i$ for any $\theta_j > \tilde{\theta}_j$. In words, given a fixed θ_i , if the share induced by θ_j belongs to i 's preferred side of the indivisibility, then so does the share induced by $\tilde{\theta}_j < \theta_j$. Now let $\Theta_i^*(\theta_j) \equiv \{\theta_i \in \Theta_i : x_i^\delta(\theta_i, \theta_j) \geq \bar{x}^{1-i}(1 - \underline{x})^i\}$ be the set of state i 's types that result in shares from the more preferable side of the indivisibility, given state j 's type. Taking the infima, we have $\theta_i^*(\theta_j) = \inf \Theta_i^*(\theta_j)$ and, noting that monotonicity of x_i^δ in θ_j gives $\Theta_i^*(\theta_j) \subseteq \Theta_i^*(\tilde{\theta}_j)$ for all $\tilde{\theta}_j < \theta_j$, we have $\theta_i^*(\theta_j) \geq \theta_i^*(\tilde{\theta}_j)$. $\square$

SI 1.8 Lemma S.6

Lemma S.6. *Fix a direct mechanism δ such that $\pi^\delta(\theta) = 0$ for all $\theta \in \Theta$ and define the bargaining residual under truth-telling by $b(\theta_0, \theta_1) \equiv u_0^\delta(\theta_0; (\theta_0, \theta_1)) - v(\theta_0, \theta_1)p$. Suppose that δ satisfies ex-post incentive compatibility, v is strictly increasing in both of its arguments, and Θ_i is a compact interval for each state i . Then, b is absolutely continuous in each argument and*

$$\frac{\partial b}{\partial \theta_i}(\theta_i, \theta_j) = \frac{\partial v}{\partial \theta_i}(\theta_i, \theta_j)(x^\delta(\theta_i, \theta_j) - p)$$

for a state i and a.e. $\theta \in \Theta$.

Proof. Fix any two types $\theta_i, \tilde{\theta}_i \in \Theta_i$ such that $\theta_i > \tilde{\theta}_i$. By always-peace and ex-post incentive

¹³ Recall that $x_0^\delta(\theta) \in [0, \underline{x}]$ implies $x_1^\delta(\theta) \in [1 - \underline{x}, 1]$ and $x_0^\delta(\theta) \in [\bar{x}, 1]$ implies $x_1^\delta(\theta) \in [0, 1 - \bar{x}]$.

compatibility, we require both

$$\begin{aligned} v(\theta_i, \theta_j)x_i^\delta(\theta_i, \theta_j) + t_i^\delta(\theta_i, \theta_j) &\geq v(\theta_i, \theta_j)x_i^\delta(\tilde{\theta}_i, \theta_j) + t_i^\delta(\tilde{\theta}_i, \theta_j) \\ v(\tilde{\theta}_i, \theta_j)x_i^\delta(\tilde{\theta}_i, \theta_j) + t_i^\delta(\tilde{\theta}_i, \theta_j) &\geq v(\tilde{\theta}_i, \theta_j)x_i^\delta(\theta_i, \theta_j) + t_i^\delta(\theta_i, \theta_j) \end{aligned}$$

for all $\theta_j \in \Theta_j$. Combining these two expressions, we have

$$(v(\theta_i, \theta_j) - v(\tilde{\theta}_i, \theta_j))x_i^\delta(\theta_i, \theta_j) \geq u_i^\delta(\theta_i; (\theta_i, \theta_j)) - u_i^\delta(\tilde{\theta}_i; (\tilde{\theta}_i, \theta_j)) \geq (v(\theta_i, \theta_j) - v(\tilde{\theta}_i, \theta_j))x_i^\delta(\tilde{\theta}_i, \theta_j).$$

Subtracting $(v(\theta_i, \theta_j) - v(\tilde{\theta}_i, \theta_j))p_i$ from all terms, we then recover bounds on the difference of the bargaining residuals $b_i(\theta_i, \theta_j) \equiv u_i^\delta(\theta_i; (\theta_i, \theta_j)) - v(\theta_i, \theta_j)p_i$ for any two types of state i given opponent j 's type,

$$(v(\theta_i, \theta_j) - v(\tilde{\theta}_i, \theta_j))(x_i^\delta(\theta_i, \theta_j) - p_i) \geq b_i(\theta_i, \theta_j) - b_i(\tilde{\theta}_i, \theta_j) \geq (v(\theta_i, \theta_j) - v(\tilde{\theta}_i, \theta_j))(x_i^\delta(\tilde{\theta}_i, \theta_j) - p_i).$$

Dividing everything by $\theta_i - \tilde{\theta}_i$ and taking the limit as $\tilde{\theta}_i \rightarrow \theta_i$, we have, by Lemma S.5 and the fact that v is C^1 , convergence to

$$\frac{\partial b_i}{\partial \theta_i}(\theta_i, \theta_j) = \frac{\partial v}{\partial \theta_i}(\theta_i, \theta_j)(x_i^\delta(\theta_i, \theta_j) - p_i)$$

for almost every $\theta \in \Theta$. Setting $i = 0$, where $b_0 = b$, establishes the claim for state 0.

To establish the condition for the opposing state $j = 1$, note that the efficiency of peace implies that total payoffs in peace always equal the stakes, $u_0^\delta(\theta_0; (\theta_0, \theta_1)) + u_1^\delta(\theta_1; (\theta_0, \theta_1)) = v(\theta_0, \theta_1)$ for all $\theta \in \Theta$. Next, we can differentiate both sides with respect to θ_1 . The right-hand side simply becomes $\partial v / \partial \theta_1$. For the left-hand side, we can put the formula just discovered to use,

$$\frac{\partial u_0^\delta}{\partial \theta_1}(\theta_0; (\theta_0, \theta_1)) = \frac{\partial v}{\partial \theta_1}(\theta_0, \theta_1) - \frac{\partial v}{\partial \theta_1}(\theta_0, \theta_1)x_1^\delta(\theta_0, \theta_1) = \frac{\partial v}{\partial \theta_1}(\theta_0, \theta_1)x^\delta(\theta_0, \theta_1).$$

Finally, because $b(\theta_0, \theta_1) = u_0^\delta(\theta_0; (\theta_0, \theta_1)) - v(\theta_0, \theta_1)p$, differentiating with respect to θ_1 recovers the stated expression for the opponent. $\square$

SI 1.9 Lemma S.7

Lemma S.7. *Fix an incentive-compatible direct mechanism δ such that $\pi^\delta(\theta) = 0$ and $x_i^\delta(\theta) = 1$ for some state i and for all $\theta \in \Theta$, and suppose that t_i^δ is integrable on $[-1, 1]^2$. Under Assumption 2, $T_i(\theta_i) \equiv E_{\theta_j}[t_i^\delta(\theta) | \theta_i]$ is constant in θ_i and $T_j(\theta_j) \equiv E_{\theta_i}[t_j^\delta(\theta) | \theta_j]$ is constant in θ_j .*

Proof. Take any incentive-compatible direct mechanism δ such that $\pi^\delta(\theta) = 0$ and $x_i^\delta(\theta) = 1$ for some state i and for all $\theta \in [-1, 1]^2$. By Assumption 2, the conditional density of θ_j given θ_i is $f(\theta_j | \theta_i; \lambda) = (1 + \lambda\theta_i\theta_j)/2$, so state i 's interim transfer from reporting $\tilde{\theta}_i$ is

$$E_{\theta_j}[t_i^\delta(\tilde{\theta}_i, \theta_j) | \theta_i] = \frac{1}{2} \int_{-1}^1 t_i^\delta(\tilde{\theta}_i, z)(1 + \lambda\theta_i z) dz = Y_i(\tilde{\theta}_i) + \lambda\theta_i Z_i(\tilde{\theta}_i),$$

with $Y_i(\theta_i) \equiv \frac{1}{2} \int_{-1}^1 t_i^\delta(\theta_i, z) dz$ and $Z_i(\theta_i) \equiv \frac{1}{2} \int_{-1}^1 z t_i^\delta(\theta_i, z) dz$. For every report $\tilde{\theta}_i \in [-1, 1]$, interim expected transfers are affine in the true type θ_i .

Because $x_i^\delta = 1$, each state's report enters its peace payoff only through the transfer. State i , holding the object, chooses a report $\tilde{\theta}_i$ that maximizes $Y_i(\tilde{\theta}_i) + \lambda\theta_i Z_i(\tilde{\theta}_i)$, whereas opponent j is compensated by the transfer alone and chooses a report $\tilde{\theta}_j$ that minimizes $E_{\theta_i}[t_i^\delta(\theta_i, \tilde{\theta}_j) | \theta_j] \equiv Y_j(\tilde{\theta}_j) + \lambda\theta_j Z_j(\tilde{\theta}_j)$ with Y_j and Z_j analogously defined by integrating state i 's transfer t_i^δ over i 's own type θ_i instead.

First, consider $\lambda = 0$. State i then maximizes $Y_i(\tilde{\theta}_i)$ over its report, so incentive-compatibility requires $Y_i(\theta_i) \geq Y_i(\tilde{\theta}_i)$ for all $\theta_i, \tilde{\theta}_i \in [-1, 1]$. Exchanging θ_i and $\tilde{\theta}_i$ gives the reverse inequality, so Y_i must be constant in type θ_i . On the other hand, opponent j minimizes $Y_j(\tilde{\theta}_j)$, and the same argument shows Y_j must be constant. Hence $T_i = Y_i$ and $T_j = -Y_j$ are constant.

Next, suppose $\lambda > 0$. Fix two types $\theta_i < \tilde{\theta}_i$ of state i . Incentive-compatibility requires that neither gain by reporting as the other, so

$$Y_i(\theta_i) + \lambda\theta_i Z_i(\theta_i) \geq Y_i(\tilde{\theta}_i) + \lambda\theta_i Z_i(\tilde{\theta}_i),$$

$$Y_i(\tilde{\theta}_i) + \lambda\tilde{\theta}_i Z_i(\tilde{\theta}_i) \geq Y_i(\theta_i) + \lambda\tilde{\theta}_i Z_i(\theta_i).$$

Adding these together, the Y_i terms cancel and the remainder factors as $\lambda(\tilde{\theta}_i - \theta_i)(Z_i(\tilde{\theta}_i) - Z_i(\theta_i)) \geq 0$. By $\lambda > 0$ and $\tilde{\theta}_i > \theta_i$, we have $Z_i(\tilde{\theta}_i) \geq Z_i(\theta_i)$, so Z_i is non-decreasing. Taking two types $\theta_j < \tilde{\theta}_j$ of

opponent j , incentive-compatibility then requires

$$\begin{aligned} Y_j(\theta_j) + \lambda\theta_j Z_j(\theta_j) &\leq Y_j(\tilde{\theta}_j) + \lambda\theta_j Z_j(\tilde{\theta}_j), \\ Y_j(\tilde{\theta}_j) + \lambda\tilde{\theta}_j Z_j(\tilde{\theta}_j) &\leq Y_j(\theta_j) + \lambda\tilde{\theta}_j Z_j(\theta_j) \end{aligned}$$

as j minimizes the expected transfer. These add to $\lambda(\tilde{\theta}_j - \theta_j)(Z_j(\tilde{\theta}_j) - Z_j(\theta_j)) \leq 0$, from which we can conclude Z_j is non-increasing.

Now, note that Z_i and Z_j share a common moment, where we can relabel the integration variables using the symmetry of the weight $z\tilde{z}$,

$$\int_{-1}^1 z Z_i(z) dz = \frac{1}{2} \int_{-1}^1 \int_{-1}^1 z \tilde{z} t_i^\delta(z, \tilde{z}) dz d\tilde{z} = \int_{-1}^1 z Z_j(z) dz. \quad (\text{S.10})$$

Because $\int_{-1}^1 z dz = 0$, for any bounded h ,

$$\int_{-1}^1 z h(z) dz = \frac{1}{4} \int_{-1}^1 \int_{-1}^1 (z - \tilde{z})(h(z) - h(\tilde{z})) dz d\tilde{z}. \quad (\text{S.11})$$

First, if h is non-decreasing, then $z \geq \tilde{z}$ implies $h(z) \geq h(\tilde{z})$ and $z \leq \tilde{z}$ implies $h(z) \leq h(\tilde{z})$, so $(z - \tilde{z})(h(z) - h(\tilde{z})) \geq 0$ and $\int_{-1}^1 z h(z) dz \geq 0$. Second, if h is non-increasing, then we have the reverse inequalities and hence $(z - \tilde{z})(h(z) - h(\tilde{z})) \leq 0$, so we can conclude $\int_{-1}^1 z h(z) dz \leq 0$. Since Z_i is non-decreasing and Z_j is non-increasing,

$$\int_{-1}^1 z Z_i(z) dz \geq 0 \geq \int_{-1}^1 z Z_j(z) dz.$$

Then by equality (S.10), each must be equal to zero.

Setting $h = Z_i$ in expression (S.11) then gives

$$\frac{1}{4} \int_{-1}^1 \int_{-1}^1 (z - \tilde{z})(Z_i(z) - Z_i(\tilde{z})) dz d\tilde{z} = \int_{-1}^1 z Z_i(z) dz = 0.$$

Because Z_i is non-decreasing, $z - \tilde{z}$ and $Z_i(z) - Z_i(\tilde{z})$ have the same sign, so the integrand is non-negative on all of $[-1, 1]^2$. A non-negative function whose integral is zero must vanish almost everywhere, so $(z - \tilde{z})(Z_i(z) - Z_i(\tilde{z})) = 0$ for almost every $(z, \tilde{z})$. Off the diagonal $z = \tilde{z}$, which has measure zero, the first factor is non-zero, so the second must vanish: $Z_i(z) = Z_i(\tilde{z})$ for almost every

pair $(z, \tilde{z})$, which holds only if Z_i is almost everywhere equal to a constant. The same argument with $h = Z_j$ shows that Z_j is almost everywhere equal to a constant. Because Z_i and Z_j are monotone, each is therefore constant on $(-1, 1)$, though their endpoint values may differ.

Restricting attention to interior types and reports, the report enters each state's objective only through Y_i and Y_j . The exchange argument of the $\lambda = 0$ case then applies unchanged: state i maximizes $Y_i(\tilde{\theta}_i)$ and opponent j minimizes $Y_j(\tilde{\theta}_j)$ over reports, so Y_i and Y_j are each constant on $(-1, 1)$.

It remains to show that the constant interior values of Z_i and Z_j are zero. Because Y_j is constant almost everywhere and $\int_{-1}^1 z dz = 0$,

$$\int_{-1}^1 Z_i(z) dz = \int_{-1}^1 \tilde{z} \left(\frac{1}{2} \int_{-1}^1 t_i^\delta(z, \tilde{z}) dz \right) d\tilde{z} = \int_{-1}^1 \tilde{z} Y_j(\tilde{z}) d\tilde{z} = 0,$$

so the constant Z_i equals zero. Analogously, because Y_i is constant almost everywhere, $\int_{-1}^1 Z_j(z) dz = \int_{-1}^1 z Y_i(z) dz = 0$, and hence $Z_j = 0$. Therefore, T_i and T_j are constant on $(-1, 1)$. Now fix an endpoint type $\theta_i \in \{-1, 1\}$. Incentive compatibility against any interior report $\tilde{\theta}_i \in (-1, 1)$ implies that $T_i(\theta_i)$ is weakly greater than the constant interior value of T_i . Conversely, for any interior type $\theta_i \in (-1, 1)$, incentive compatibility against an endpoint report $\tilde{\theta}_i \in \{-1, 1\}$ implies $T_i(\theta_i) \geq Y_i(\tilde{\theta}_i) + \lambda \theta_i Z_i(\tilde{\theta}_i)$. Taking $\theta_i \rightarrow \tilde{\theta}_i$ shows that the constant interior value is weakly greater than $Y_i(\tilde{\theta}_i) + \lambda \tilde{\theta}_i Z_i(\tilde{\theta}_i) = T_i(\tilde{\theta}_i)$. Thus, T_i is constant on $[-1, 1]$. The analogous argument establishes that T_j is constant on $[-1, 1]$, completing the proof. $\square$

SI 1.10 Lemma S.8

Lemma S.8. *Under Assumptions 2 and 3, $V_i(\theta_i; \lambda)$ is strictly increasing in θ_i for every $\lambda \in [0, 1]$.*

Proof. Fix λ and let $\theta_i < \tilde{\theta}_i$. By Assumption 2, $V_i(\theta_i; \lambda) = \int_{-1}^1 \phi(\theta_i + z) f(z | \theta_i; \lambda) dz$. Then, by adding and subtracting $\int_{-1}^1 \phi(\tilde{\theta}_i + z) f(z | \theta_i; \lambda) dz$,

$$\begin{aligned} V_i(\tilde{\theta}_i; \lambda) - V_i(\theta_i; \lambda) &= \int_{-1}^1 \phi(\tilde{\theta}_i + z) f(z | \tilde{\theta}_i; \lambda) dz - \int_{-1}^1 \phi(\theta_i + z) f(z | \theta_i; \lambda) dz \\ &= \int_{-1}^1 [\phi(\tilde{\theta}_i + z) - \phi(\theta_i + z)] f(z | \theta_i; \lambda) dz + \int_{-1}^1 \phi(\tilde{\theta}_i + z) [f(z | \tilde{\theta}_i; \lambda) - f(z | \theta_i; \lambda)] dz. \end{aligned}$$

On the first integral, $\phi(\tilde{\theta}_i + z) - \phi(\theta_i + z) > 0$ for all $z \in [-1, 1]$, since $\tilde{\theta}_i > \theta_i$ and ϕ is strictly

increasing by Assumption 3. As $f(\cdot | \theta_i; \lambda)$ is a probability density, it follows that the first term is strictly positive.

Next, note that $f(\theta_j | \tilde{\theta}_i; \lambda) - f(\theta_j | \theta_i; \lambda) = \lambda(\tilde{\theta}_i - \theta_i)\theta_j/2$ by Assumption 2, so we can rewrite the second integral as

$$\frac{\lambda(\tilde{\theta}_i - \theta_i)}{2} \int_{-1}^1 z \phi(\tilde{\theta}_i + z) dz = \frac{\lambda(\tilde{\theta}_i - \theta_i)}{2} \int_{-1}^1 z [\phi(\tilde{\theta}_i + z) - \phi(\tilde{\theta}_i)] dz,$$

where the constant $\phi(\tilde{\theta}_i)$ may be inserted because $\int_{-1}^1 z dz = 0$. By ϕ strictly increasing, the term $\phi(\tilde{\theta}_i + z) - \phi(\tilde{\theta}_i)$ has the same sign as z , so the integrand $z[\phi(\tilde{\theta}_i + z) - \phi(\tilde{\theta}_i)]$ is non-negative. Together with $\lambda(\tilde{\theta}_i - \theta_i)/2 \geq 0$, the second term is non-negative.

Adding the two, the first integral is strictly positive and the second non-negative, so $V_i(\tilde{\theta}_i; \lambda) > V_i(\theta_i; \lambda)$. The statement for V_j is symmetric. $\square$

SI 1.11 Lemma S.9

Lemma S.9. *Suppose power p and the costs of war (c_0, c_1) are common knowledge, but there are uncertain stakes v . If Assumptions 2 and 3 hold and*

$$c_0 + c_1 < p_j(V_j(1; \lambda) - V_i(-1; \lambda)) \quad (5)$$

for a state i and opponent j , then there does not exist an incentive-compatible direct mechanism δ such that, for all $\theta \in \Theta$, $\pi = 0$ and $x_i = 1$ for all $(\pi, x, t) \in \text{supp}(\delta(\theta))$. Therefore, there does not exist a transfer-protocol crisis bargaining game that admits an always-peaceful equilibrium.

Proof. Fix a state i and opponent j , and let Assumptions 2 and 3 and inequality (5) hold. Suppose for contradiction that there exists an incentive-compatible direct mechanism δ such that $\pi = 0$ and $x_i = 1$ for all $(\pi, x, t) \in \text{supp}(\delta(\theta))$, for all $\theta \in \Theta$. By Lemma S.7, the interim transfer is constant in each state's own type, so δ pays opponent j some constant transfer t .

Because state i keeps the object, this settlement gives state i of type θ_i the interim expected payoff $V_i(\theta_i; \lambda) - t$ and opponent j the payoff t . On the other hand, war yields interim expected payoffs of $V_i(\theta_i; \lambda)p_i - c_i$ and $V_j(\theta_j; \lambda)p_j - c_j$ for state i and j , respectively. A state accepts the settlement only if it is at least as good as war, hence $V_i(\theta_i; \lambda) - t \geq V_i(\theta_i; \lambda)p_i - c_i$ for state i and

$t \geq V_j(\theta_j; \lambda)p_j - c_j$ for opponent j . Rearranging state i 's condition and invoking the fact that $p_j = 1 - p_i$ yields $t \leq V_i(\theta_i; \lambda)p_j + c_i$.

By Lemma S.8, $V_j(\theta_j; \lambda)$ is strictly increasing in θ_j , and hence the bound $V_j(\theta_j; \lambda)p_j - c_j$ is largest at $\theta_j = 1$. The opponent j 's condition therefore holds for every θ_j if and only if $t \geq V_j(1; \lambda)p_j - c_j$. Likewise, $V_i(\theta_i; \lambda)p_j + c_i$ is strictly increasing in θ_i and hence smallest at $\theta_i = -1$, making state i 's condition hold for every θ_i if and only if $t \leq V_i(-1; \lambda)p_j + c_i$. A constant transfer thus sustains always-peace if and only if

$$V_j(1; \lambda)p_j - c_j \leq t \leq V_i(-1; \lambda)p_j + c_i.$$

Such a t exists if and only if this interval is non-empty, requiring

$$c_0 + c_1 \geq p_j (V_j(1; \lambda) - V_i(-1; \lambda)),$$

a contradiction, concluding the proof. □

SI 1.12 Lemma S.10

Lemma S.10. *Under Assumptions 2 and 3, $V_j(1; \lambda) - V_i(-1; \lambda)$ is strictly increasing in λ*

Proof. By Assumption 2, the conditional density of θ_i given $\theta_j = 1$ is $f(\theta_i | 1; \lambda) = (1 + \lambda\theta_i)/2$, so

$$V_j(1; \lambda) = \frac{1}{2} \int_{-1}^1 \phi(1 + z)(1 + \lambda z) dz,$$

which is affine in λ . Differentiating,

$$\frac{d}{d\lambda} V_j(1; \lambda) = \frac{1}{2} \int_{-1}^1 z \phi(1 + z) dz = \frac{1}{2} \int_{-1}^1 z [\phi(1 + z) - \phi(1)] dz,$$

where the constant $\phi(1)$ may be inserted because $\int_{-1}^1 z dz = 0$. Because ϕ is strictly increasing, $\phi(1 + z) - \phi(1)$ has the same sign as z , so the integrand is non-negative and strictly positive for $z \neq 0$, and hence $\frac{d}{d\lambda} V_j(1; \lambda) > 0$. By the same argument, with $f(\theta_j | -1; \lambda) = (1 - \lambda\theta_j)/2$,

$$\frac{d}{d\lambda} V_i(-1; \lambda) = -\frac{1}{2} \int_{-1}^1 z [\phi(z - 1) - \phi(-1)] dz < 0.$$

$\theta_0 \sim F_0$		
$\theta_1 \sim F_1$	(θ_0, θ_1)	
	fight	settle
	fight	$w_0(\theta), w_1(\theta)$
	settle	$w_0(\theta), w_1(\theta)$
		$v(\theta)x, v(\theta)(1-x)$

Figure 6: Example of a share-protocol crisis bargaining game with uncertain stakes. State 0 (row) and 1 (column) simultaneously choose to fight or settle. Dashed boxes reflect state uncertainty about their opponent's type. Fighting yields war payoffs, whereas settling returns a share of the object corresponding to a fixed division x .

Therefore, $V_j(1; \lambda) - V_i(-1; \lambda)$ is strictly increasing in λ . $\square$

SI 2 Share protocols

Consider a subclass of crisis bargaining games in which settlements may only divide or allocate shares of the object, precluding an exchange of transfers between states in the form of side payments, compromises on other issues, or other external compensation mechanisms. I refer to this subclass as *share-protocol* crisis bargaining games.

Definition SI 2.1 (Share Protocol). *A crisis bargaining game Γ uses a share protocol if and only if, for every $a \in A$ and all $(\pi, x, t) \in \text{supp}(\gamma(a))$, $t = 0$ whenever $\pi = 0$.*

To see that always-peaceful equilibria can exist in this subclass, consider an example of a simple share-protocol crisis bargaining game shown in Figure 6, in which states observe their private types and simultaneously choose whether to fight or settle. They receive their war payoffs $w_i(\theta)$ unless they both settle, in which case state 0 receives an x share of $v(\theta)$ whereas state 1 receives the complementary $1 - x$ share, for a fixed $x \in [0, 1]$. Always-peace requires an x large enough to guarantee every type of state 0 their expected war payoff, but small enough to guarantee every type of state 1 their expected war payoff. Because expected war payoffs increase with expected stakes, this can be satisfied by a settlement x such that $c_1/\sup_{\theta_1} V_1(\theta_1) > x - p > -c_0/\sup_{\theta_0} V_0(\theta_0)$. A peaceful settlement is generally possible; indeed, peace is guaranteed if shares are always distributed according to the known balance of power, $x = p$. This simple example admits an always-peaceful equilibrium, so share-protocol crisis bargaining games that admit always-peaceful equilibria exist.

SI 2.1 Proposition S.1

Proposition S.1. *There exists a share-protocol game with uncertain stakes that admits an always-peaceful equilibrium. Formally, suppose power p and the costs of war (c_0, c_1) are common knowledge, but there are uncertain stakes v . Then, there exists an incentive-compatible direct mechanism δ such that, for all $\theta \in \Theta$, $\pi = 0$ and $t = 0$ for all $(\pi, x, t) \in \text{supp}(\delta(\theta))$.*

Proof. Consider the direct mechanism δ defined by $\delta(\theta)(\{(0, p, 0)\}) = 1$ for all $\theta \in \Theta$. Fix a state i , true type θ_i , and report $\tilde{\theta}_i$. For every $\theta_j \in \Theta_j$, $u_i^\delta(\tilde{\theta}_i; (\theta_i, \theta_j)) = v(\theta_i, \theta_j)p_i$ and hence $U_i^\delta(\tilde{\theta}_i; \theta_i) = E_{\theta_j}[v(\theta_i, \theta_j)p_i | \theta_i] = V_i(\theta_i)p_i$, which does not vary in the report $\tilde{\theta}_i$ and hence satisfies the incentive-compatibility constraint. Moreover, because $c_i > 0$ for each state i , it is clear to see the interim participation constraint is also satisfied, $U_i^\delta(\theta_i; \theta_i) > V_i(\theta_i)p_i - c_i$.

By Lemma S.1, there exists a crisis bargaining game with a Bayesian Nash equilibrium of which δ is an equivalent direct mechanism. Because $t = 0$ for all $(\pi, x, t) \in \text{supp}(\delta(\theta))$ for all $\theta \in \Theta$, and choosing off-path transfers such that $t = 0$, every $a \in A$ satisfies $t = 0$ for all $(\pi, x, t) \in \text{supp}(\gamma(a))$. Therefore, the constructed game belongs to the share-protocol subclass. $\square$

Always-peace can be achieved despite uncertain stakes. A natural way for this to occur is if a protocol can be instituted to allocate shares of the object in proportion to the known distribution of power between states. In this way, there is a clear similarity between the case of uncertain stakes under a share-protocol and the case of uncertainty about the costs of war (e.g., Proposition 3 of Fey and Ramsay, 2011). Because power is commonly known, this division is certain to leave both states better off than war, regardless of what the true stakes are. Moreover, because the share-protocol restriction concerns transfers and not the divisions of the object, the proposition also establishes that uncertain stakes alone do not preclude always-peace. It is the interaction with an indivisibility, and not the uncertainty itself, that the impossibility results identify.

A wider range of always-peaceful divisions can usually be facilitated with uncertain stakes than with uncertain costs of war. In the example, peace does not require the settlement to exactly match the power balance; any division in a range satisfying the most demanding type of each state can work. For any game in the share-protocol subclass, the interim participation constraint from Lemma S.2 requires that each state expect to receive at least as much value from their share as they would from war, $E_{\theta_j}[v(\theta)x_i^\delta(\theta) | \theta_i] \geq V_i(\theta_i)p_i - c_i$ for all $\theta_i \in \Theta_i$.

This condition is not sufficient, however, and incentive compatibility sharply restricts how shares vary with reports. Indeed, *none* of the variation in a state's interim expected utility across types can come from changes in the equilibrium share allocation. The variation is carried by two channels: a direct channel through the stakes of the dispute, and an indirect channel through beliefs about an opponent's private information. On the direct channel, each side can expect a higher utility when their type increases, simply because they expect the dispute to involve a more valuable object. Less straightforwardly, a marginal increase in a state's type can also change their beliefs about their opponent's type, which also affects their expectation of the object's value. Whether this serves to push a state's interim expected utility up or down depends on the sign of the correlation between private information. Accordingly, the indirect channel vanishes under independent types. Corollary S.1 (SI 1.3) presents the formal statement.

The constraint has substantive implications for the kinds of share protocols that can generate always-peace. Suppose a rule awarded a larger share to reports of higher expected stakes. A share of the object is worth more to every type, so every type would claim the higher expectations, and the rule would unravel. Without transfers, a state that reports higher expected stakes than it holds gives up nothing in return for the larger share. A rule that responds to reports can be incentive compatible only if a report that raises a state's share also changes how that share varies with the opponent's type, so that no type expects to gain by misreporting. The constant settlement at the power-balanced share instead makes private information irrelevant and satisfies every type on its own. Thus, while Proposition S.1 demonstrates that always-peace is possible in a share-protocol crisis bargaining game, the possibility rests on rules that ignore reports rather than respond to them.

SI 3 Proofs

This section presents the proofs for formal statements presented in the main article. I provide a restatement of the result for convenience.

SI 3.1 Proposition B.1

Proposition B.1. *Suppose the object's value is common to both states and commonly known, $v_0 = v_1 = v$, while power p and the costs of war (c_0, c_1) may be unknown. Fix any nonempty set of*

feasible divisions $X \subseteq [0, 1]$ and any incentive-compatible direct mechanism δ . Then there exists an incentive-compatible direct mechanism $\tilde{\delta}$ such that $x \in X$ for all $(\pi, x, t) \in \text{supp}(\tilde{\delta}(\theta))$ and all $\theta \in \Theta$, with the same probability of war and the same ex-post payoffs as δ at every type profile. Moreover, $\tilde{\delta}$ is robustly implementable whenever δ is, and if $U_i^\delta(\theta_i; \theta_i) \geq E_{\theta_j}[w_i(\theta) | \theta_i]$ for each state i and all $\theta_i \in \Theta_i$, there exists a crisis bargaining game whose feasible divisions lie in X with an equilibrium of which $\tilde{\delta}$ is an equivalent direct mechanism.

Proof. Fix any $\hat{x} \in X$ and define $\tilde{\delta}$ from δ realization by realization. Whenever $\delta(\theta)$ realizes a war outcome, let $\tilde{\delta}(\theta)$ realize the same outcome. Whenever $\delta(\theta)$ realizes a peaceful outcome $(0, x, t)$, let $\tilde{\delta}(\theta)$ realize $(0, \hat{x}, t + v(x - \hat{x}))$. The map is affine in the outcome and hence measurable, so $\tilde{\delta}(\theta) \in \Delta(\Omega)$ for all θ , and we maintain $t_0 = -t_1$ because $v(x_0 - \hat{x}_0) + v(x_1 - \hat{x}_1) = v[(x_0 + x_1) - (\hat{x}_0 + \hat{x}_1)] = 0$. By construction, every realized peaceful share lies in X and $\pi^{\tilde{\delta}}(\theta) = \pi^\delta(\theta)$ for all θ .

Moreover, payoffs are unchanged at every realization: for each state i , we have

$$v\hat{x}_i + t_i + v(x_i - \hat{x}_i) = vx_i + t_i.$$

Because the stakes are known, the adjustment is computed at the constant rate v and does not depend on the reported or true types, so the equality holds at every report. War outcomes are unchanged by construction, and therefore $u_i^{\tilde{\delta}}(\tilde{\theta}_i; \theta) = u_i^\delta(\tilde{\theta}_i; \theta)$ for each state i , all reports $\tilde{\theta}_i$, and all $\theta \in \Theta$, which implies $U_i^{\tilde{\delta}}(\tilde{\theta}_i; \theta_i) = U_i^\delta(\tilde{\theta}_i; \theta_i)$ everywhere as well.

Each of (IC), (EPIC), and (EPVA) is a family of inequalities in these objects, so each holds for $\tilde{\delta}$ if and only if it holds for δ . If δ is deterministic, $\tilde{\delta}$ maps its single outcome to a single outcome and is deterministic, so $\tilde{\delta}$ is robustly implementable whenever δ is. Finally, if δ satisfies the interim participation constraint, then so does $\tilde{\delta}$, and Lemma S.1 yields a crisis bargaining game and equilibrium of which $\tilde{\delta}$ is an equivalent direct mechanism, with off-path settlements chosen in X . Therefore, every feasible division of the constructed game lies in X . $\square$

SI 3.2 Proposition 1

Proposition 1. *When the stakes of the dispute v are known, no restriction on the feasible divisions of the object changes the probabilities of war or the payoffs attainable across crisis bargaining games, even as uncertainty about power or the costs of war persists.*

Proof. The result follows immediately by Proposition B.1. $\square$

SI 3.3 Theorem 1

Theorem 1. *No crisis bargaining game with uncertain stakes and a boundary indivisibility admits an always-peaceful equilibrium unless the costs of war are sufficiently large. Formally, suppose power p and the costs of war (c_0, c_1) are common knowledge, but there are uncertain stakes v . If Assumption 1 holds and there is a boundary indivisibility with $\bar{x} \in (0, 1]$ such that*

$$c_0 + c_1 < (\bar{x} - p_i)(\bar{V}_j - \underline{V}_i) - (1 - \bar{x})(Ev - \underline{V}_j) \quad (3)$$

for a state i and opponent j , then there does not exist an incentive-compatible direct mechanism δ such that, for all $\theta \in \Theta$, $\pi = 0$ and $x_i \geq \bar{x}$ for all $(\pi, x, t) \in \text{supp}(\delta(\theta))$.

Proof. Fix a state i and opponent j . Suppose for contradiction that inequality (3) holds and some crisis bargaining game with the boundary indivisibility admits an always-peaceful equilibrium σ^* . By Remark A.2, there exists an incentive-compatible equivalent direct mechanism δ such that, for all $\theta \in \Theta$, $\pi = 0$ and $x_i \geq \bar{x}$ for all $(\pi, x, t) \in \text{supp}(\delta(\theta))$. Since every realized outcome in the support of $\delta(\theta)$ satisfies $x_i \geq \bar{x}$, the induced expected peaceful share also satisfies $x_i^\delta(\theta) \geq \bar{x}$ for all $\theta \in \Theta$.

First, we can place a lower bound on $EU_j^{\sigma^*} \equiv E_{\theta_j} U_j^{\sigma^*}(\theta_j)$. By Lemma S.2 and the definition of equivalent direct mechanism, $U_j^{\sigma^*}(\bar{\theta}_j) \geq \bar{V}_j p_j - c_j$. Additionally, by Lemma S.4, we know that $U_j^{\sigma^*}(\bar{\theta}_j) - U_j^{\sigma^*}(\theta_j) \leq (1 - \bar{x})(\bar{V}_j - \underline{V}_j)$. Combining the two inequalities and rearranging,

$$\begin{aligned} U_j^{\sigma^*}(\theta_j) &\geq U_j^{\sigma^*}(\bar{\theta}_j) - (1 - \bar{x})(\bar{V}_j - \underline{V}_j) \\ &\geq \bar{V}_j p_j - c_j - (1 - \bar{x})(\bar{V}_j - \underline{V}_j) = (\bar{x} - p_i)\bar{V}_j + (1 - \bar{x})\underline{V}_j - c_j. \end{aligned} \quad (\text{S.12})$$

By Lemma S.4, $U_j^{\sigma^*}(\theta_j) \geq U_j^{\sigma^*}(\bar{\theta}_j)$ for all $\theta_j \in \Theta_j$, and hence $EU_j^{\sigma^*} \geq (\bar{x} - p_i)\bar{V}_j + (1 - \bar{x})\underline{V}_j - c_j$.

Next, we can establish an upper bound on $EU_i^{\sigma^*}$. By Lemma S.4, we have that $U_i^{\sigma^*}(\theta_i) \geq U_i^{\sigma^*}(\underline{\theta}_i) + \bar{x}(V_i(\theta_i) - \underline{V}_i)$ for all $\theta_i \in \Theta_i$. Taking expectations over $\theta_i \sim F_i$, we can then recover $EU_i^{\sigma^*} \geq U_i(\theta_i) + \bar{x}(E_{\theta_i} V_i(\theta_i) - \underline{V}_i)$. By Assumption 1 and Lemma S.2, we have $EU_i^{\sigma^*} \geq \bar{x}Ev - (\bar{x} - p_i)\underline{V}_i - c_i$.

Because $EU_i^{\sigma^*} = Ev - EU_j^{\sigma^*}$ for any always-peaceful equilibrium σ^* , we can deduce that

$$EU_j^{\sigma^*} = Ev - EU_i^{\sigma^*} \leq (1 - \bar{x})Ev + (\bar{x} - p_i)V_i + c_i. \quad (\text{S.13})$$

Together, bounds (S.12) and (S.13) imply $c_i + c_j \geq (\bar{x} - p_i)(\bar{V}_j - V_i) - (1 - \bar{x})(Ev - V_j)$, a contradiction to inequality (3). $\square$

SI 3.4 Proposition 2

Proposition 2. *No transfer-protocol game with uncertain stakes admits an always-peaceful equilibrium unless the costs of war are sufficiently large. Formally, suppose power p and the costs of war (c_0, c_1) are common knowledge, but there are uncertain stakes v . If Assumption 1 holds and*

$$c_0 + c_1 < p_j(\bar{V}_j - V_i)$$

for a state i and opponent j , then there does not exist an incentive-compatible direct mechanism δ such that, for all $\theta \in \Theta$, $\pi = 0$ and $x_i = 1$ for all $(\pi, x, t) \in \text{supp}(\delta(\theta))$.

Proof. Setting $\bar{x} = 1$, the result follows immediately by Theorem 1. $\square$

SI 3.5 Proposition 3

Proposition 3. *For any interior indivisibility, there exists a crisis bargaining game with uncertain stakes that admits an always-peaceful equilibrium. Formally, suppose power p and the costs of war (c_0, c_1) are common knowledge, but there is uncertainty about stakes v . If there is an interior indivisibility given by any $\underline{x} \in (0, 1)$ and $\bar{x} \in (\underline{x}, 1)$, then there exists an incentive-compatible direct mechanism δ such that $\pi = 0$ and $x \notin (\underline{x}, \bar{x})$ for all $(\pi, x, t) \in \text{supp}(\delta(\theta))$, for all $\theta \in \Theta$.*

Proof. First, for any interior indivisibility such that $p \notin (\underline{x}, \bar{x})$, the direct mechanism δ such that $\delta(\theta)(\{(0, p, 0)\}) = 1$ for all $\theta \in \Theta$ satisfies the statement by Proposition S.1, as its only settlement avoids $(\underline{x}, \bar{x})$. In the game constructed by Lemma S.1, give each state two additional actions, L_i and H_i . Let (L_0, L_1) implement $(0, \underline{x}, 0)$, let (H_0, H_1) implement $(0, \bar{x}, 0)$, and let every other action profile involving one of these additional actions produce war. A unilateral deviation to an additional action therefore yields war, so the equilibrium is unchanged. Both $\underline{x}$ and $\bar{x}$ are feasible, and no

peaceful share lies in $(\underline{x}, \bar{x})$. Therefore, the constructed game has the specified interior indivisibility and admits an always-peaceful equilibrium. Now suppose $p \in (\underline{x}, \bar{x})$.

For proof by example, conjecture the following direct mechanism. For any reported types $\tilde{\theta}$, the mechanism designer assigns the outcome

$$\delta(\tilde{\theta}) = \begin{cases} (0, \bar{x}, -v(\tilde{\theta})(\bar{x} - p)) & \text{with prob. } q \\ (0, \underline{x}, v(\tilde{\theta})(p - \underline{x})) & \text{with prob. } 1 - q. \end{cases}$$

State 0 with type θ_0 then has an interim expected utility for reporting as type $\tilde{\theta}_0$ of

$$U_0^\delta(\tilde{\theta}_0; \theta_0) = V_0(\theta_0)(q\bar{x} + (1 - q)\underline{x}) - E_{\theta_1}[v(\tilde{\theta}_0, \theta_1) | \theta_0](q(\bar{x} - p) - (1 - q)(p - \underline{x})). \quad (\text{S.14})$$

Further, let $q = (p - \underline{x})/(\bar{x} - \underline{x})$ so that $q\bar{x} + (1 - q)\underline{x} = p$ and $q(\bar{x} - p) - (1 - q)(p - \underline{x}) = 0$. Then, the expression (S.14) reduces to $U_0^\delta(\tilde{\theta}_0; \theta_0) = V_0(\theta_0)p_0$, independent of the report. First, the direct mechanism satisfies the interim participation constraint, as $V_0(\theta_0)p_0 \geq V_0(\theta_0)p_0 - c_0$. Second, the direct mechanism trivially satisfies the incentive-compatibility constraint, as the interim expected utility does not depend on the reported type.

Similarly, the interim expected utility for state 1 of type θ_1 reporting as type $\tilde{\theta}_1$ simplifies to $U_1^\delta(\tilde{\theta}_1; \theta_1) = V_1(\theta_1)p_1$ and hence an analogous argument follows for state 1.

By Lemma S.1, there exists a crisis bargaining game with a Bayesian Nash equilibrium of which δ is an equivalent direct mechanism. Because $x \notin (\underline{x}, \bar{x})$ for all $(\pi, x, t) \in \text{supp}(\delta(\theta))$ for all $\theta \in \Theta$, and choosing off-path settlements so that $x \notin (\underline{x}, \bar{x})$ without loss of generality, every $a \in A$ satisfies $x \notin (\underline{x}, \bar{x})$ for all $(\pi, x, t) \in \text{supp}(\gamma(a))$. Therefore, the constructed game has the interior indivisibility given by $(\underline{x}, \bar{x})$. $\square$

SI 3.6 Theorem 2

Theorem 2. *No crisis bargaining game with uncertain stakes and an interior indivisibility admits a robust always-peaceful equilibrium unless the costs of war are sufficiently large. Formally, suppose power p and the costs of war (c_0, c_1) are common knowledge, but there are uncertain stakes v . Further, let v be strictly increasing in both of its arguments and Θ_i be a compact interval for each state i .*

Then, if there is an interior indivisibility given by $\underline{x} \in (0, 1)$ and $\bar{x} \in (\underline{x}, 1)$ such that

$$c_0 + c_1 < \frac{(\bar{x} - p)(p - \underline{x})}{\bar{x} - \underline{x}} \max_i \left\{ \max_{\theta_j} [v(\bar{\theta}_i, \theta_j) - v(\underline{\theta}_i, \theta_j)] \right\}, \quad (4)$$

there does not exist a robustly implementable direct mechanism δ such that $\pi = 0$ and $x \notin (\underline{x}, \bar{x})$ for all $(\pi, x, t) \in \text{supp}(\delta(\theta))$, for all $\theta \in \Theta$.

Proof. Suppose, for contradiction, there exists such a robustly implementable direct mechanism δ . Recall b as the bargaining residual defined by Lemma S.6. Ex-post voluntary agreements requires $u_i^\delta(\theta_i; (\theta_i, \theta_j)) \geq v(\theta_i, \theta_j)p_i - c_i$ for each state i . Combining these, we recover a bound on the bargaining residual, $c_1 \geq b(\theta) \geq -c_0$ for all $\theta \in \Theta$.

By Lemma S.5, there is a threshold $\theta_i^*(\theta_j) \in \Theta_i$ that partitions state i 's types into settlements with shares below $\underline{x}$ and above $\bar{x}$. Further, let $v_i^*(\theta_j) \equiv v(\theta_i^*(\theta_j), \theta_j)$ be the value of the stakes at state i 's threshold and $b_i^*(\theta_j) \equiv b(\theta_i^*(\theta_j), \theta_j)$ the bargaining residual at state i 's threshold, given any opponent type θ_j .

Now fix $\theta_1 \in \Theta_1$. Then, by Lemma S.6,

$$\frac{\partial b}{\partial \theta_0}(\theta_0, \theta_1) = \frac{\partial v}{\partial \theta_0}(\theta_0, \theta_1)(x^\delta(\theta_0, \theta_1) - p) \leq -\frac{\partial v}{\partial \theta_0}(\theta_0, \theta_1)(p - \underline{x})$$

for any $\theta_0 < \theta_0^*(\theta_1)$. Because b is absolutely continuous, by integrating from $\underline{\theta}_0$ to $\theta_0^*(\theta_1)$, we have

$$b(\underline{\theta}_0, \theta_1) - b_0^*(\theta_1) \geq (p - \underline{x})(v_0^*(\theta_1) - v(\underline{\theta}_0, \theta_1)).$$

Likewise, we also have

$$\frac{\partial b}{\partial \theta_0}(\theta_0, \theta_1) = \frac{\partial v}{\partial \theta_0}(\theta_0, \theta_1)(x^\delta(\theta_0, \theta_1) - p) \geq \frac{\partial v}{\partial \theta_0}(\theta_0, \theta_1)(\bar{x} - p) \quad (\text{S.15})$$

by Lemma S.6, for any $\theta_0 > \theta_0^*(\theta_1)$. Integrating again, this time from $\theta_0^*(\theta_1)$ to $\bar{\theta}_0$, yields

$$b(\bar{\theta}_0, \theta_1) - b_0^*(\theta_1) \geq (\bar{x} - p)(v(\bar{\theta}_0, \theta_1) - v_0^*(\theta_1)). \quad (\text{S.16})$$

Combining the two integrated inequalities with the bound on the residual b , we establish a bound

on the stakes of the dispute at state 0's threshold,

$$v(\underline{\theta}_0, \theta_1) + \frac{c_0 + c_1}{p - \underline{x}} \geq v_0^*(\theta_1) \geq v(\bar{\theta}_0, \theta_1) - \frac{c_0 + c_1}{\bar{x} - p}$$

for all types $\theta_1 \in \Theta_1$. Rearranging, we require that

$$c_0 + c_1 \geq \frac{(\bar{x} - p)(p - \underline{x})}{\bar{x} - \underline{x}} (v(\bar{\theta}_0, \theta_1) - v(\underline{\theta}_0, \theta_1)) \quad (\text{S.17})$$

for every $\theta_1 \in \Theta_1$.

State 1 follows by symmetric argument. Fixing $\theta_0 \in \Theta_0$, we can again apply Lemma S.6 to bound the partial derivative of b ,

$$\frac{\partial b}{\partial \theta_1}(\theta_0, \theta_1) \leq -\frac{\partial v}{\partial \theta_1}(\theta_0, \theta_1)(p - \underline{x}); \quad \theta_1 > \theta_1^*(\theta_0) \quad (\text{S.18})$$

$$\frac{\partial b}{\partial \theta_1}(\theta_0, \theta_1) \geq \frac{\partial v}{\partial \theta_1}(\theta_0, \theta_1)(\bar{x} - p); \quad \theta_1 < \theta_1^*(\theta_0). \quad (\text{S.19})$$

Integrating inequality (S.18) from $\theta_1^*(\theta_0)$ to $\bar{\theta}_1$ and inequality (S.19) from $\underline{\theta}_1$ to $\theta_1^*(\theta_0)$ in the same way recovers, respectively,

$$b_1^*(\theta_0) - b(\theta_0, \bar{\theta}_1) \geq (p - \underline{x})(v(\theta_0, \bar{\theta}_1) - v_1^*(\theta_0)) \quad (\text{S.20})$$

$$b_1^*(\theta_0) - b(\theta_0, \underline{\theta}_1) \geq (\bar{x} - p)(v_1^*(\theta_0) - v(\theta_0, \underline{\theta}_1)). \quad (\text{S.21})$$

Together with the residual bound, inequalities (S.20) and (S.21) imply

$$c_0 + c_1 \geq \frac{(\bar{x} - p)(p - \underline{x})}{\bar{x} - \underline{x}} (v(\theta_0, \bar{\theta}_1) - v(\theta_0, \underline{\theta}_1)) \quad (\text{S.22})$$

for every $\theta_0 \in \Theta_0$.

Both (S.17) and (S.22) therefore hold for every $\theta_1 \in \Theta_1$ and $\theta_0 \in \Theta_0$, respectively, contradicting inequality (4). $\square$

SI 3.7 Proposition 4

Proposition 4. *Take any transfer-protocol crisis bargaining game with uncertainty about stakes, and let Assumptions 2 and 3 hold. If inequality (5) holds at $\lambda \in [0, 1]$ but not at $\tilde{\lambda} \in [0, 1]$, then it must be that $\lambda > \tilde{\lambda}$, and that inequality (5) holds at every $\lambda' > \lambda$ and does not hold at any $\lambda'' < \tilde{\lambda}$.*

Proof. By Lemma S.10, $V_j(1; \lambda) - V_i(-1; \lambda)$ is strictly increasing in λ , and hence the threshold $p_j(V_j(1; \lambda) - V_i(-1; \lambda))$ is also strictly increasing in λ . Because (5) holds at λ but not at $\tilde{\lambda}$,

$$p_j(V_j(1; \tilde{\lambda}) - V_i(-1; \tilde{\lambda})) \leq c_0 + c_1 < p_j(V_j(1; \lambda) - V_i(-1; \lambda)),$$

which implies $\lambda > \tilde{\lambda}$.

Now, choose any $\lambda' > \lambda$. The threshold strictly increases, which implies

$$c_0 + c_1 < p_j(V_j(1; \lambda) - V_i(-1; \lambda)) < p_j(V_j(1; \lambda') - V_i(-1; \lambda')),$$

and hence inequality (5) holds at λ' . Next, choose any $\lambda'' < \tilde{\lambda}$. The threshold strictly decreases, which implies

$$c_0 + c_1 \geq p_j(V_j(1; \tilde{\lambda}) - V_i(-1; \tilde{\lambda})) > p_j(V_j(1; \lambda'') - V_i(-1; \lambda'')),$$

and hence inequality (5) does not hold at λ'' . □

Supplemental Information References

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